



Spatially Intelligent Digital Twin Capabilities and Characteristics

A Digital Twin Consortium Whitepaper

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1 INTRODUCTION

Digital twins accelerate decision-making by increasing productivity, optimizing efficiencies, and improving outcomes across industries. Digital twins enable situational awareness, allowing better predicting and forecasting of future outcomes, helping businesses produce more cost-efficient and effective products and services.

To realize this transformative value, digital twin systems need to be designed and architected with best practices for scalability, interoperability, and composability. This starts with defining the capabilities of a given digital twin. By defining these capabilities, digital twin solutions can be developed, tested, and deployed.

Core to digital twins is the *virtual representation* of reality, including the facilities, assets, and real-world conditions being modeled and managed. Representing, visualizing, modeling and analyzing the *location and spatial (positional) relationships* between these entities is essential.

The fundamental principle of a spatially intelligent digital twin lies in its ability to process and interpret spatial relationships between physical assets, infrastructure, and operational systems., a spatially intelligent digital twin utilizes location intelligence, geospatial context, and depending on the use case, may include real-time motion tracking into its framework. Example use cases include smart cities, logistics, energy, manufacturing, and disaster management.

Dr. Werner Kuhn states “*Location* is the base concept of all spatial information, enabling *where* questions. The concept comes in two commonly used forms: location as a spatial relation between what is located and what locates it; and location as a region in the world defined by such relations.”¹

As described above, a spatially intelligent digital twin is based on the concept of location. This paper will explore these locational characteristics in the context of its capabilities.

1.1 PURPOSE AND INTENDED AUDIENCE

This document was prepared by the Digital Twin Consortium’s Architecture, Engineering, Construction, and Operations (AECO) Working Group². The examples provided within this document focus on built infrastructure, building, and facility use cases. However, the concepts described within can apply to a wide spectrum of digital twin use cases, industries and disciplines.

¹ Kuhn, W. (2012). Core concepts of spatial information for transdisciplinary research. *International Journal of Geographical Information Science*, 26(12), 2267–2276.

² Digital Twin Consortium *AECO Working Group*.

Spatially Intelligent Digital Twin Capabilities and Characteristics

This document provides a guide for the identification and documentation of locational capabilities and characteristics for the purposes of establishing digital twin requirements and design.

This document will help organizations:

1. Document the capabilities and resulting value streams provided through the ability to visualize, understand, and analyze the locational characteristics of real-world entities and conditions.
2. Understand the distinction between different forms of locational representations including geometric (3D models), spatial, and geospatial models.
3. Document the key characteristics of locational representations in a digital twin, so that organizations can capture locational characteristics in a consistent, common fashion, thereby enabling digital twin system-to-system integration.
4. Capture the spatially intelligent digital twin's locational characteristics in the context of capabilities using the Digital Twin Consortium's Capabilities Periodic Table (CPT).³

This document can be used by business executives; enterprise, business, and solution architects; system designers, and developers to understand and describe the relevance of location to digital twin capabilities.

1.2 DOCUMENT ORGANIZATION

This document is organized as follows:

- Section 2 introduces digital twins, digital twin virtual representations, and the value and importance of location in digital twin virtual representations.
- Section 3 introduces the concept of capabilities, the Digital Twin Consortium's Capabilities Periodic Table (CPT), how the CPT can be used to define capabilities for a given digital twin use case, and how locational capabilities can be described in the context of the CPT.
- Section 3 helps organizations define what location intelligence *capabilities* a digital twin must deliver.
- Section 4 introduces different locational models and the different characteristics (attributes) that are used to describe the locational information in a digital twin, based on the definition of digital twin capabilities.

Section 4 also helps organizations to understand the specifications and *characteristics* of their location-related data, to deliver location intelligence capabilities.

³ Digital Twin Consortium *Capabilities Periodic Table*.

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- Section 5 provides guidance on alternative methods that organizations can use to document their locational data and location intelligence capabilities and characteristics for digital twins.

1.3 METHODOLOGY

This document can be used as a high-level user guide for considering, selecting, and documenting locational capabilities and characteristics of digital twins. The following overall approach is recommended:

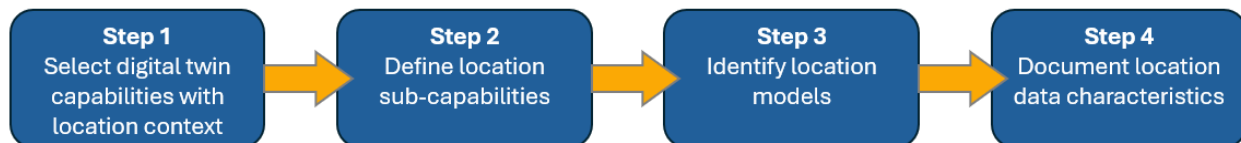


Figure 1-1: Approach for defining digital twin locational capabilities and characteristics.

1. Step 1. Select digital twin capabilities needing a location context – The Digital Twin Consortium provides the Digital Twin CPT to help organizations select and describe their digital twin capability requirements. The CPT is described in Sections 3.3 and 3.4.
2. Step 2. Define location sub-capabilities – Location awareness will be an essential enhancement of many capabilities on the CPT for the spatially Intelligent digital twin. The CPT is designed to enable hierarchical, increasingly detailed, descriptions of digital twin capabilities. Section 3.5 provides an approach, with examples, for documenting the location-enabled digital twin sub-capabilities.
3. Step 3. Identify locational models – Once the locational capability requirements for a digital twin have been selected and described, organizations can then determine the types of locational data (“locational data models”) that best enable and satisfy those capability requirements. Section 4.1 introduces and describes key locational models including 3D geometric models, spatial, and geospatial data types.
4. Step 4. Document location data characteristics – The properties of required locational data must be defined, once the appropriate locational data models are selected. Section 4.2 introduces the key characteristics and properties of spatial data that should be evaluated, selected, and described.

By completing Steps 1-4, organizations should have a structured, well-defined description of locational capability and data requirements for their digital twins. This information can then be used to help design, develop, and operate digital twins to best meet organizational needs and to provide business value.

1.4 LOCATION AS A DIGITAL TWIN CAPABILITY

Core to all digital twins is a virtual representation of the real world — a digital representation of the real-world objects, entities, facilities, and conditions being modeled in the twin. For many

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digital twin use cases this includes understanding the location of the assets being represented. The question “where” is at the heart of many digital twins: Where are the rooms in the building? Where are individual buildings related to the larger site and context? Where are the buildings, roads, and other assets in the city, and how do they connect and relate to each other?

Location is therefore essential to understand the layout and context of the real-world entities being represented. Location intelligence — the insight gained from visualizing and analyzing data in a locational context — is an essential capability to realize value from digital twins.

Regardless of the scope and scale of a digital twin or what lifecycle phase(s) the twin is focused on — planning, design, construction, or operations — location matters. Many digital twins are not possible, or are certainly less capable, without location awareness.

In the context of digital twins, location extends beyond geo-coordinates and includes essential relative positioning details that drive deeper insights and operational relevance. Rather than focusing solely on latitude and longitude, which might hold value for specific components like pumps or turbines, understanding “relative location” allows digital twins to capture the spatial relationships critical to functional performance and system interaction.

For instance, in an industrial setup, knowing how far a pump is from other assets, such as tanks, turbines, or safety equipment, helps predict performance, streamline maintenance, and optimize operational efficiency.

This broader, relational view of location unlocks richer insights for common digital twin applications and provides a foundational layer for predictive analytics, enabling these models to not only represent individual assets but also their contextual dependencies and influence on surrounding components.

Emphasizing relative location empowers digital twin strategies to effectively replicate the dynamic interactions across complex systems, ensuring that insights derived are both accurate and actionable.

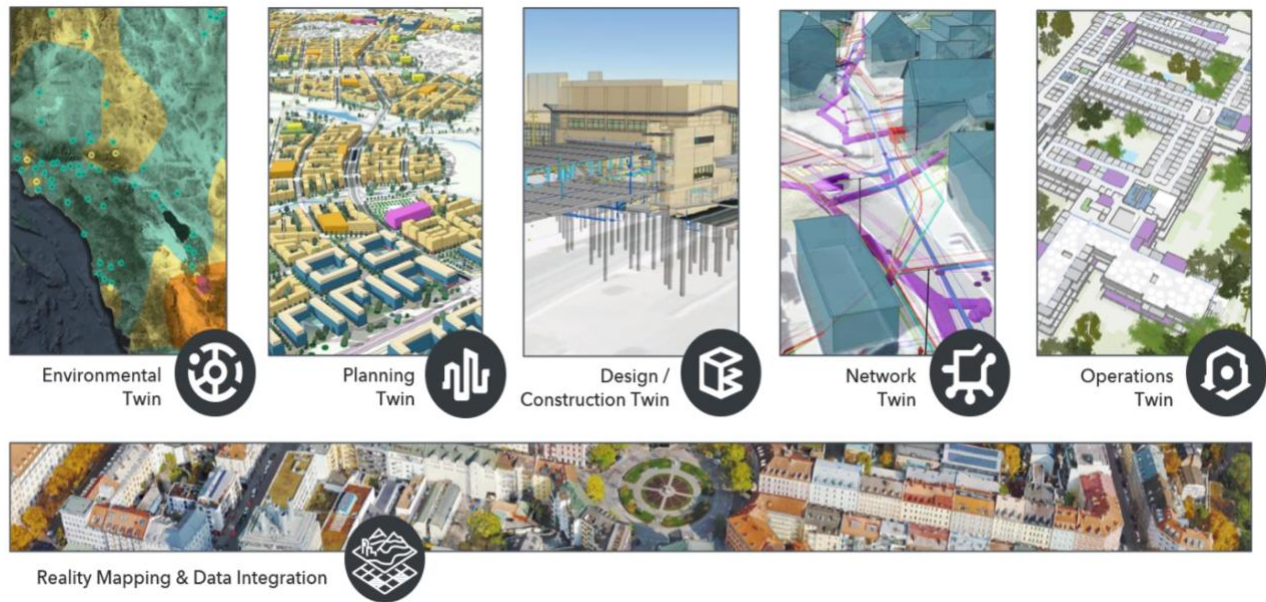


Figure 1-2: Location as a key enabler of digital twins.⁴

2 DIGITAL TWINS, VIRTUAL REPRESENTATIONS, AND LOCATION

A digital twin is a virtual representation of real-world entities and processes, synchronized at a specified frequency and fidelity. Digital twins use real-time and historical data to represent the past and present and simulate predicted futures. Digital twins are motivated by outcomes, tailored to use cases, powered by integration, built on data, guided by domain knowledge, and implemented in information and operational technology (IT/OT) systems.

Table 2-1 shows the criteria by which an IT/OT solution can be considered a digital twin or a digital twin system of systems, as described in the Digital Twin Consortium's Digital Twin Platform Stack Architectural Framework⁵.

⁴ Esri. Digital Twin Overview. Retrieved November 2024 from <https://www.esri.com/en-us/digital-twin/overview>

⁵ Digital Twin Consortium. July 2023. *Platform Stack Architectural Framework: An Introductory Guide*.

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Criterion	Model	Simulation	Dashboard (or other Application)	Digital Twin	Digital Twin System
Virtual (Stored) Representation	✓			✓	✓
Virtual (Computational) Representation / Simulation & Forecasting		✓		✓	✓
Built on Data (Synchronization for Initialization)			✓	✓	✓
Synchronized (including Feedback & Control)			✓	✓	✓
System Integration			✓		✓

Table 2-1: Classification of models and simulations through to digital twins and digital twin systems.⁶

2.1 DIGITAL TWIN REPRESENTATIONS

As described further in the Digital Twin Consortium's Digital Twin Platform Stack Architectural Framework, digital twins include both:

1. Virtual (stored) representations – A model consisting of stored digital information representing past and current states of real-world assets, entities, and conditions. This includes tabular data representing operational conditions of the real-world assets as well as a visual representation of the assets themselves. The visual representation of assets in a digital twin are inherently locational (geometric, spatial, or geospatial) in nature. A stored representation of a building in a digital twin is shown in Figure 2-1.

⁶ Digital Twin Consortium, *Platform Stack Architectural Framework: An Introductory Guide*.



Figure 2-1: Stored representation of a building in a digital twin.⁷

2. Virtual (computational) representations – Derived or calculated data, generated from one or more executable algorithms including one or more functions, using stored representation data as inputs to simulate or forecast alternative or future conditions. Computational representations provide the analytical and “intelligence” capabilities within a digital twin to answer questions and to inform decision-making activities. Stored representation data act as the inputs to computational representations, as shown in Figure 2-2.

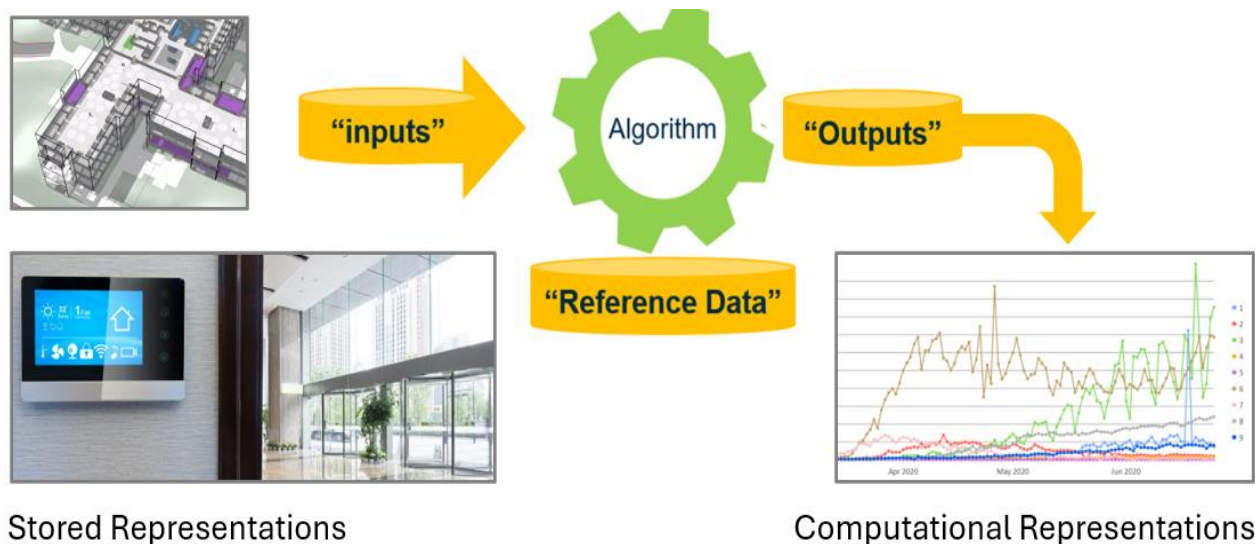


Figure 2-2: Computational representation in a digital twin.

⁷ Esri. ArcGIS Indoors Overview. Retrieved November 2024 from <https://www.esri.com/en-us/arcgis/products/arcgis-indoors>

2.2 REPRESENTING REAL WORLD OBJECTS IN A DIGITAL TWIN

Real world objects are included in stored representations in a digital twin. The method and characteristics by which the objects are depicted and described vary based on the size and scale of the object being represented, as shown in Figure 2-3.



Figure 2-3: Representing real world objects in a digital twin stored representation.

1. Individual Asset — Individual equipment items such as a turbine in a pump station, a jet engine on a plane, or a robotic armature performing an assembly function in a manufacturing facility. A logical schematic of the asset may be sufficient for visual representation in a digital twin; the exact location of the asset may not be important. For example, a digital twin might depict the logical connection between important components within a turbine. The basic connective properties are most important; how components connect within the turbine may be more important than depicting the exact shape or location of the object.
2. Individual building, facility, or system — Basic spatial properties such as the layout of the building, facility, or system become important. Spatial relationships such as adjacency, connectivity, and distance with the facility become important. For example, it may be relevant to know what rooms in a building are adjacent to other rooms, how far apart two rooms are from each other, and how the two rooms are connected by hallways or other circulation areas.
3. Campus, city, region, or large complex systems — A digital twin can encompass and represent a larger region such as a campus, a town or city, or a larger region. At this scale, understanding geographic concepts including size and extent, proximity, connectivity, overlap, movement between locations, and the presence of temporal spatial patterns become increasingly important.

2.3 SPATIAL INTELLIGENCE PROVIDING LOCATIONAL RELEVANCE TO DIGITAL TWINS

In all three cases shown in Figure 2-3, location – including geometric shape, spatial, and geospatial properties – is essential to the representations of real-world objects.

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As the scale of a digital twin increases from individual assets to individual facilities and up to entire cities, and as the scope of a digital twin expands to more complex systems, the ability to understand spatial and geographic properties and relationships becomes an important capability. Answering “Spatially intelligent” questions such as the following are key enablers to understand the real world, predict and optimize future states, and to satisfy the business objectives of a digital twin:

- What assets are near other assets?
- What assets are physically connected?
- How do people and vehicles travel between assets?
- Where are food sources in relation to people?
- Where is housing in relation to people?
- What are the current conditions at a given set of locations, or within a set of defined assets?

Figure 2-4 illustrates how digital twins can answer the latter question. The following illustrates a digital twin of a building showing current temperatures and energy usage within rooms in a building. Locational data is essential to enable this viewing and analysis capability.



Figure 2-4: Digital twins requiring locational information.⁸

⁸ Image from Di Zio, Jessica. (25 Feb 2021). *Learn How to Integrate IoT Data into Your Model with New Data Visualization Extensions*. Autodesk.

3 DESCRIBING THE SPATIALLY INTELLIGENT DIGITAL TWIN LOCATIONAL CAPABILITIES

Organizations must first start by identifying and describing the capabilities their digital twins will provide – what their digital twins will do. This includes the visual and locational capabilities including the ability to visual real-world assets and conditions in a virtual representation and analytical functions that consider space and geography in computational representations.

Defining the locational capabilities of the Spatially Intelligent digital twin is an essential precursor to describing the locational characteristics of the informational assets.

This describes how organizations can describe these locational capabilities of their digital twins. Chapter 4 then describes how to document a digital twin’s locational characteristics, based on the required capabilities.

3.1 INTRODUCTION TO CAPABILITIES

A *capability* is the ability or capacity that a business may possess (or require) to achieve a specific purpose or outcome. A capability statement describes *what* a business does, independently of *where, why, or how* that capability is achieved⁹. Capabilities provide:

- A common vocabulary for all stakeholders to agree on what a business (or digital twin) does,
- A way to see what is common amongst digital twins in a system-of-systems,
- The basis for defining and prioritizing design requirements and investments, and
- People, processes, and (technology) solutions are how capabilities are delivered.

Documenting required capabilities is an essential precursor for digital twin design, solution identification, development, and deployment.

Capabilities are often expressed in terms of “the ability to...”. For example, the capability to perform data analysis might be described as “*the ability to analyze data to gain insights and inform decisions.*” And then, a requirement for this capability would be stated as, “our digital twin *requires the ability to perform data analysis.*”

Capabilities can be decomposed as required to clearly articulate what a business (or a digital twin) does to the level of granularity required for planning, design, and implementation. The highest level capabilities are commonly referred to as “level 1” capabilities, with further breakdown of those capabilities referred to as “level 2”, “level 3”, etc.

⁹ Business Architecture Guild. (2024). Business Architecture Body of Knowledge (BizBok). Version 13.

3.2 CAPABILITY REFERENCE MODELS

A capability reference model, alternately referred to as a “capability map” is a means by which an enterprise or system’s capabilities can be organized, logically grouped, described, and evaluated in a structured and hierarchical fashion.

The Digital Twin Consortium provides the Capabilities Periodic Table as an adaptable and scalable capability reference model applicable to all digital twins.

3.3 THE DIGITAL TWIN CAPABILITIES PERIODIC TABLE

Digital twins can enable and deliver a broad range of capabilities. The Capabilities Periodic Table was developed to describe and document the breadth and scope of digital twin capabilities.

The CPT is an architecture- and technology agnostic capability reference model used for digital twin requirements definition. It is aimed at organizations that want to design, develop, deploy, and operate digital twins based on use case capability.

The CPT uses a periodic table approach with capabilities grouped or “clustered” around common characteristics. The visual and hierarchical nature of the CPT enables understanding and use by, and between, a broad range of stakeholders from senior executives to business and solution architects, as well as those that will use and benefit from use of a digital twin.

The CPT is organized into six logical groupings, matching capabilities that have similar characteristics and application, as shown in Figure 3-1.

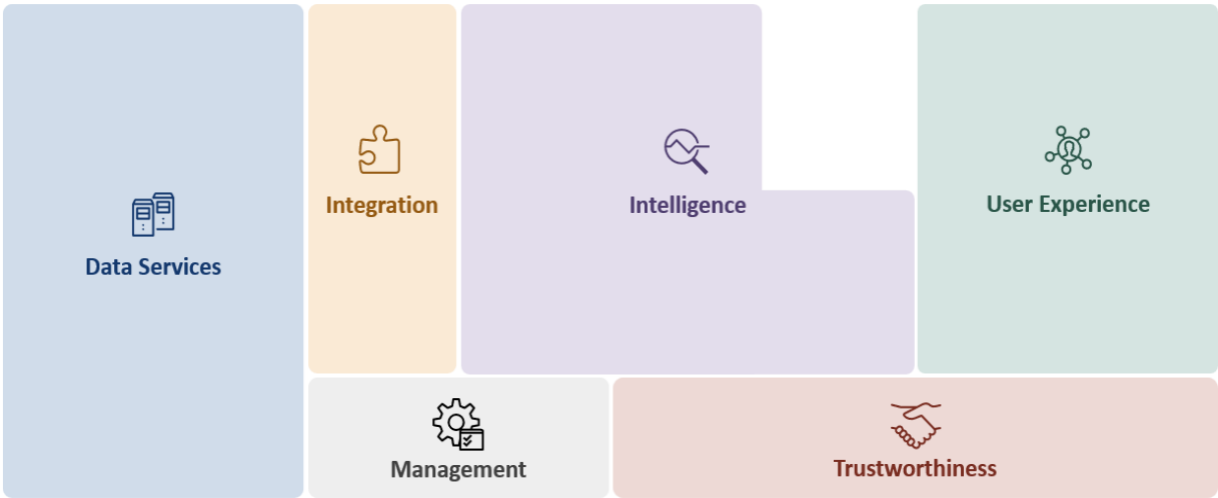


Figure 3-1: Digital twin capabilities periodic table categories.¹⁰

Table 3-1 describes each grouping, or functional categorization of capabilities, on the CPT.

¹⁰ See the Digital Twin Consortium website for more detailed information on the *Capabilities Periodic Table*.

No	Category Name	Category Description
DS	Data Services	Enables data access, ingestion and data management across the platform from the edge to the cloud. It establishes the physical to virtual connection and receives data directly from equipment sensors or control systems, performs localized processing and distributes to other tiers.
IR	Integration	Enables data access to existing internal and external enterprise systems and applications. Enables communication across different digital twins.
IC	Intelligence	Provides an environment for the development and deployment of industrial Digital Twin solutions. It provides the services for data integration, basic and advanced analytics, AI, orchestration and other Digital Twin process capabilities.
UX	User Experience	Provides the user with the ability to interact with Digital Twins and visualize its data.
MN	Management	System and ecosystem management capabilities.
TW	Trustworthiness	Security, privacy, safety, reliability and resilience capabilities.

Table 3-1: Digital twin capabilities periodic table category descriptions¹¹.

3.3.1 LEVEL 1 AND LEVEL 2 CAPABILITY HIERARCHIES

Figure 3-2 shows the level 1 capabilities of digital twins, as defined in the CPT.

These “level 1” capabilities are organized under the six categories shown above and can be further decomposed to level 2 sub-capabilities that define and describe each level 1 capability at a more granular level.¹²

3.4 USING THE CPT TO DEVELOP DIGITAL TWIN SCOPE AND REQUIREMENTS

The CPT can be used to develop the scope and requirements for specific digital twin use cases. The specific level 1 capabilities that apply to a given digital twin scope can be selected from the

¹¹ Digital Twin Consortium. *Capabilities Periodic Table*.

¹² Level 2 sub-categorization has not yet been developed. However, the Digital Twin Consortium provides guidance for organizations to develop level 2 sub-capabilities.

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CPT, whether the digital twin is for a manufacturing facility, an operational wind farm, an electrical utility network, or for a newly constructed office building.

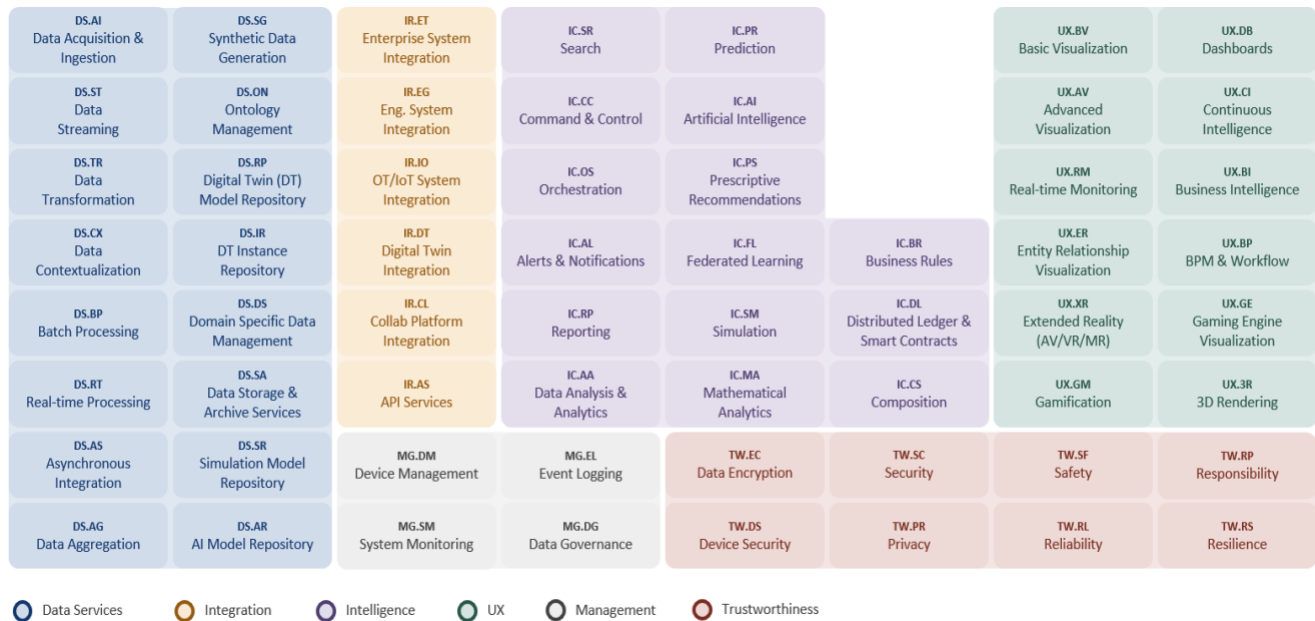


Figure 3-2: Digital twin capabilities (level 1)¹³.

3.4.1 SELECTING LEVEL 1 CAPABILITIES

Organizations should start by identifying the level 1 capabilities required by their digital twin. The selected capabilities are the foundation for requirements definition for a given use case, and the basis for solution design.

The process to identify the capabilities for a given digital twin use case starts with “down-selection” of the full set of level 1 digital twin capabilities to just those level 1 capabilities relevant to the given use case¹⁴.

Figure 3-3 illustrates the subset of level 1 capabilities that might apply to a digital twin for condition monitoring and maintenance of an operational wind farm.

¹³ Digital Twin Consortium. *Capabilities Periodic Table*.

¹⁴ Further details on the capability filtering and selection process is provided in the *Digital Twin Consortium Capabilities Periodic Table User's Guide*.

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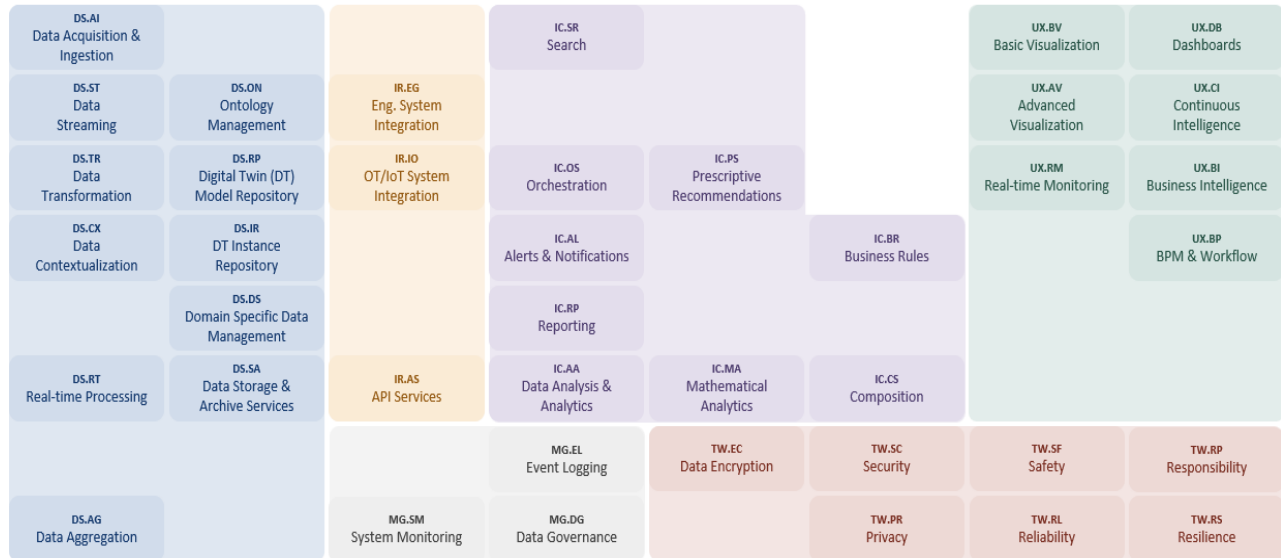


Figure 3-3: Selecting capabilities (Level 1) for a digital twin¹⁵.

3.4.2 DESCRIBING CAPABILITIES

After selecting the level 1 capabilities selected for a given use case, the characteristics and attributes of those capabilities should be described. The Digital Twin Consortium provides templates to capture the descriptive details of each capability, as shown in Figure 3-4. However, any format, template, or tool can be used to describe the selected CPT capabilities for a given digital twin use case.

Digital Twin Ecosystem Capabilities Periodic Table							
USE CASE		Wind Farm Energy Prediction					
What it is		Use Case Application					
No	Digital Twin Capability	Category	Use Case Capability Requirements	Attributes	Edge	Cloud	Level
1	Data Acquisition and Ingestion	1. Data Services	Ingest real-time sensor readings with high frequency and low latency	50 tags at 5 second frequency with 10kb payload per message for 10,000 wind turbines	Full	x	Advanced
2	Data Streaming	1. Data Services			x	x	Nice to have
3	Data Transformation and Wrangling	1. Data Services			x	x	
4	Data Contextualization	1. Data Services			x	x	

Figure 3-4: Describing the capabilities (Level 1) for a digital twin¹⁶.

3.4.3 DEFINING LEVEL 2+ DIGITAL TWIN CAPABILITIES

The Digital Twin Consortium does not prescribe or provide level 2 capabilities for the CPT.

¹⁵ Digital Twin Consortium. *Capabilities Periodic Table*.

¹⁶ Digital Twin Consortium. *Capabilities Periodic Table*.

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Organizations may develop level 2+ sub-capabilities for each level 1 capability as needed, and in a fashion that best fits the organization's needs and use cases.

Figure 3-5 illustrates possible level 2 and 3 sub-capabilities for the level 1 capability *IC.AI, Artificial Intelligence*. The level 1 capability is decomposed into four level 2 sub-capabilities. One of those level 2 capabilities, *IC.AI.GA, Generative AI*, is then decomposed into five level 3 capabilities.

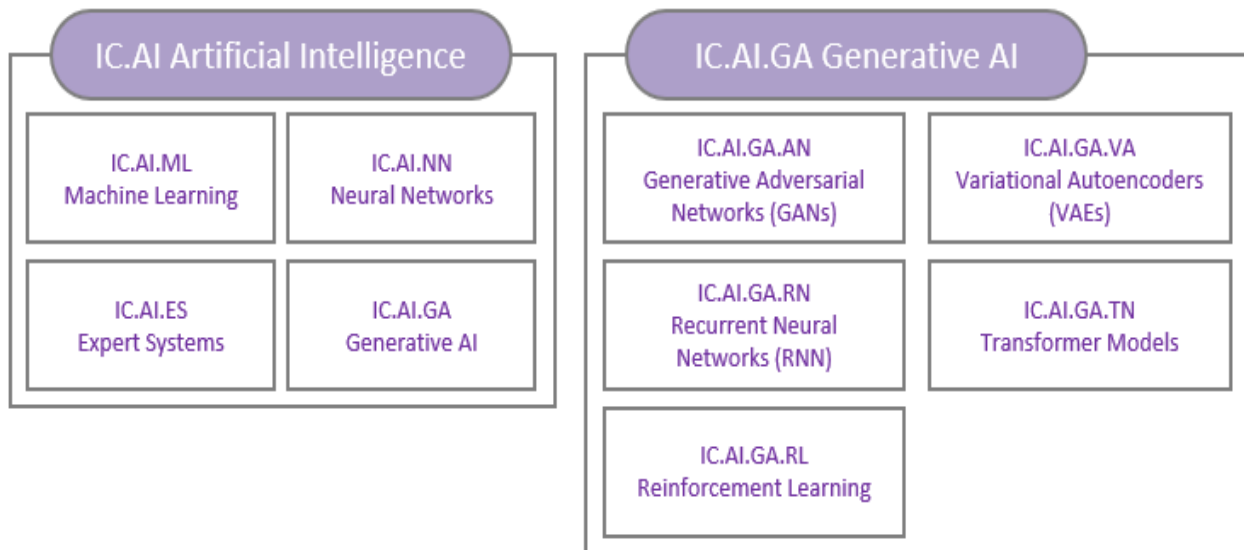


Figure 3-5: Defining level 2+ capabilities for a digital twin.

3.5 DESCRIBING DIGITAL TWIN LOCATIONAL CAPABILITIES

The level 1 capability *DS.DS, Domain-Specific Data Management* (highlighted in Figure 3-6) is the “core” capability to which the management of geospatial (and temporal) data is aligned.

- A set of level 2+ capabilities can be defined to describe the location data management lifecycle core functions - acquisition, creation, storage, editing, maintenance, archiving, and deleting.

Level 2 capabilities can be defined for this and other level 1 capabilities to describe specific location intelligence (visualization, analytical) abilities and functions.

- For example, a set of level 2 capabilities can be developed that describe the use of interactive maps as part of the *UX.BV, Basic Visualization* and *UX.AV, Advanced Visualization* level 1 capabilities.
- In another example, the level 1 capability *IC.AA, Data Analytics and Analysis*, can be decomposed into a set of level 2 capabilities describing the ability to analyze data in a digital twin based on proximity, patterns of distribution, and movement of assets.

These two examples are explored in more detail in Section 3.5.1.

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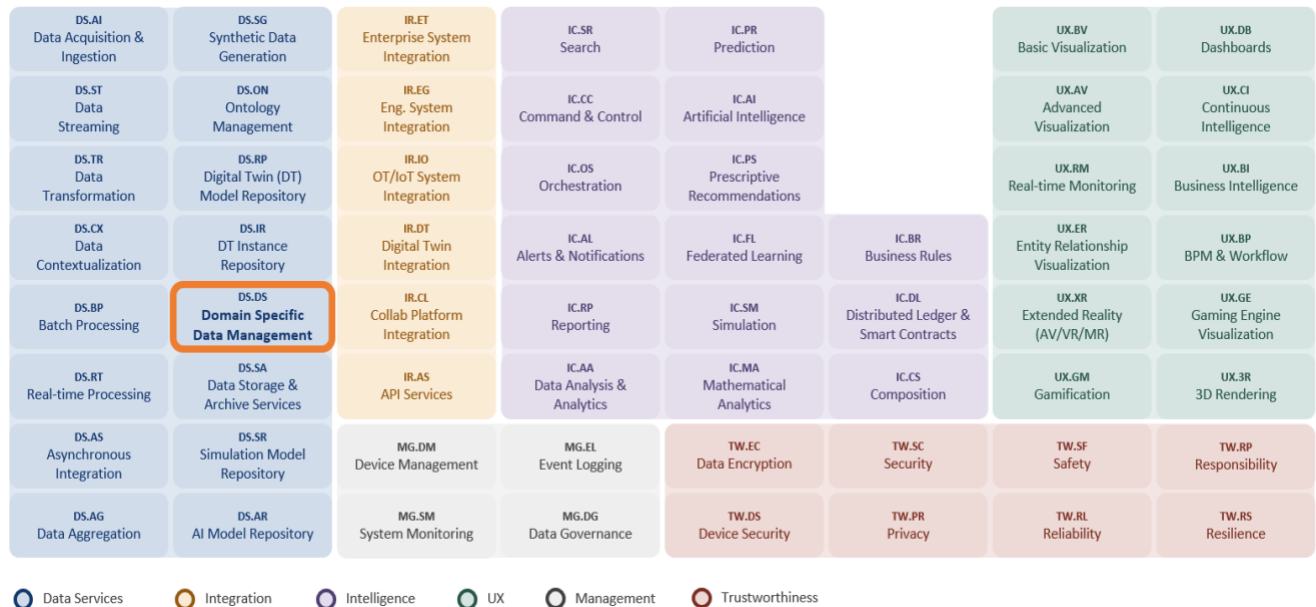


Figure 3-6: Level 1 capabilities for management of location data.¹⁷

3.5.1 DEFINING LOCATIONAL CAPABILITIES – ILLUSTRATIVE EXAMPLES

Two examples of defining (Level 2) locational capabilities are provided, one for a digital twin basic visualization use case and one for a digital twin data analytics analysis use case:

USE CASE 1 – BASIC VISUALIZATION

1. The CPT Level 1 capability UV.BV, Basic Visualization provides digital twins with the ability to help people understand the significance of real-world data and conditions by portraying that information in a visual context.
2. This can include portrayal of data on a map display – A digital twin may require the ability to include an interactive digital map to portray the location of assets, asset operational status, and current conditions. In Figure 3-7, a digital twin interface includes a map display of road assets and current traffic conditions.

¹⁷ Digital Twin Consortium. *Capabilities Periodic Table*.

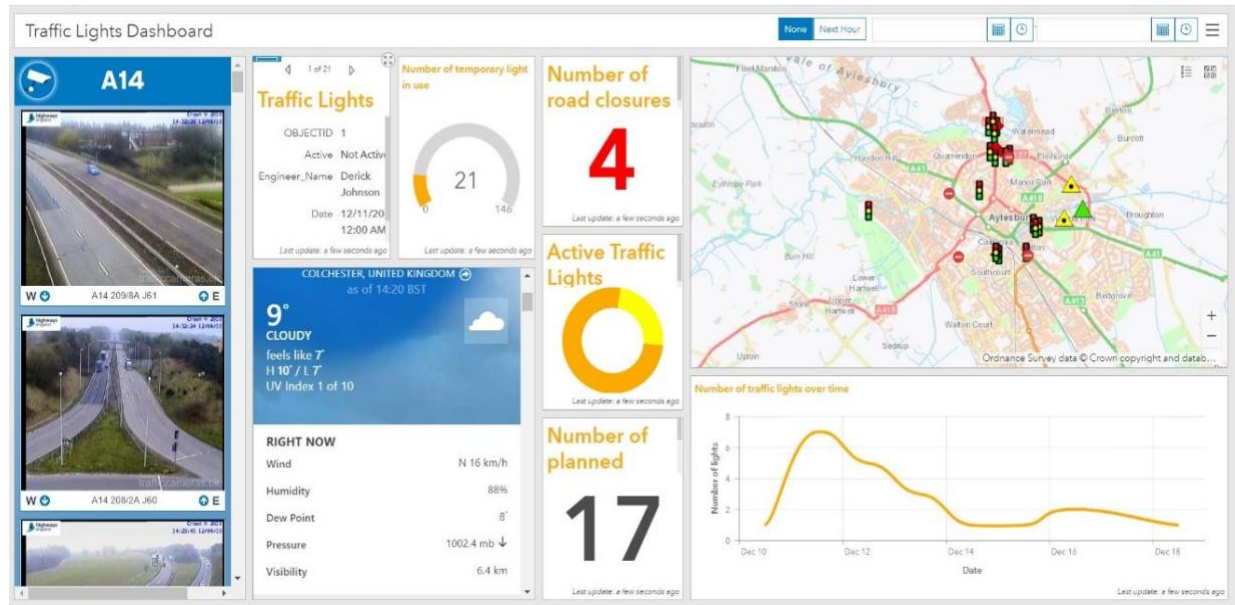


Figure 3-7: Digital twin basic visualization with map displays.¹⁸

- Level 2 Capabilities for Basic Visualization on a Map Display may include the following. For illustrative purposes, the Level 2 capabilities shown in Figure 3-8 are adapted from the University Consortium for Geographic Information Science (UCGIS) *Geographic Information Science and Technology Body of Knowledge* (GIS&T BOK)¹⁹, but organizations may reference other frameworks, or establish their own, as needed.

¹⁸ Esri ArcGIS Blog. Transportation Articles. Retrieved March 2024 from <https://www.esri.com/arcgis-blog/products/arcgis-solutions/transportation/>

¹⁹ The UCGIS GIS&T BOK provides a structured framework for documenting the domain of geographic information science and its associated technologies. The GIS&T BOK was last updated in January 2024. <https://www.ucgis.org/gis-t-body-of-knowledge>

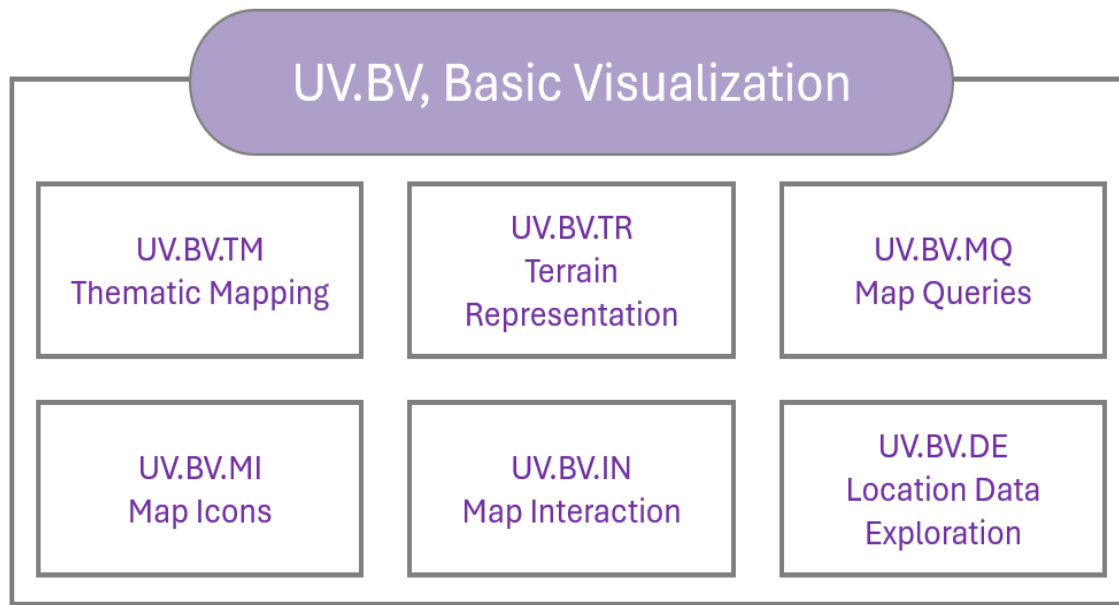


Figure 3-8: Level 2 capabilities for digital twin basic visualization with map displays.

These Level 2 capabilities might be described as follows:

“Our digital twin requires the ability to...”

- *UV.BV.TM, Thematic Mapping* – ... portray real-world assets, current asset characteristics, and operational status and condition using varying colors and symbols, with data aggregated into thematic categories on an interactive and dynamic map display,
- *UV.BV.MI, Map Icons* – ... portray assets in a unique fashion using unique symbols for different asset types on an interactive and dynamic map display,
- *UV.BV.TR, Terrain Representation* – ... represent the real-world situation and environment in which our assets, asset operational status and conditions are located, using terrain and/or imagery base map data,
- *UV.BV.IN, Map Interaction* – ... allow the user to interactively move around the map display, using map zoom and pan techniques, and to toggle map content (layers) on and off, to tailor the map display to their visualization preferences,
- *UV.BV.MQ, Map Queries* – ... allow the user to select and highlight one or more assets or conditions on the map based on a selection or filtering criteria. All assets meeting the user-entered selection (filtering) criteria will then be visually highlighted on the map display,
- *UV.BV.DE, Location Data Exploration* – ... allow the user to point at (select) any given asset on the map display and view and explore the information (attributes and other resources, such as stored digital pictures, etc.) associated with the selected asset.

USE CASE 2 – DATA ANALYTICS AND ANALYSIS

1. The CPT Level 1 capability *IC.AA, Data Analysis and Analytics* provides digital twins with the ability to understand past trends and predict future trends and conditions using historic and other types of data (including synthetic data²⁰).
2. This can include analysis of data based on spatial relationships – A digital twin may require the ability to analyze the spatial (locational) relationships between assets (e.g., clustering or dispersion of vehicles) or movement of assets (e.g., movement of vehicles on a road network). In Figure 3-9, a digital twin provides the ability to calculate spatial statistics across a given region (e.g., a city, county, or other administrative unit) based on sensor or other real world data feeds.

²⁰ Synthetic data are created using generative artificial intelligence (AI), by applying a sampling technique to real-world data, or by creating simulation scenarios where models and processes interact to create completely new data not directly taken from the real world. Adapted from:

<https://www.gartner.com/en/information-technology/glossary/synthetic-data>.

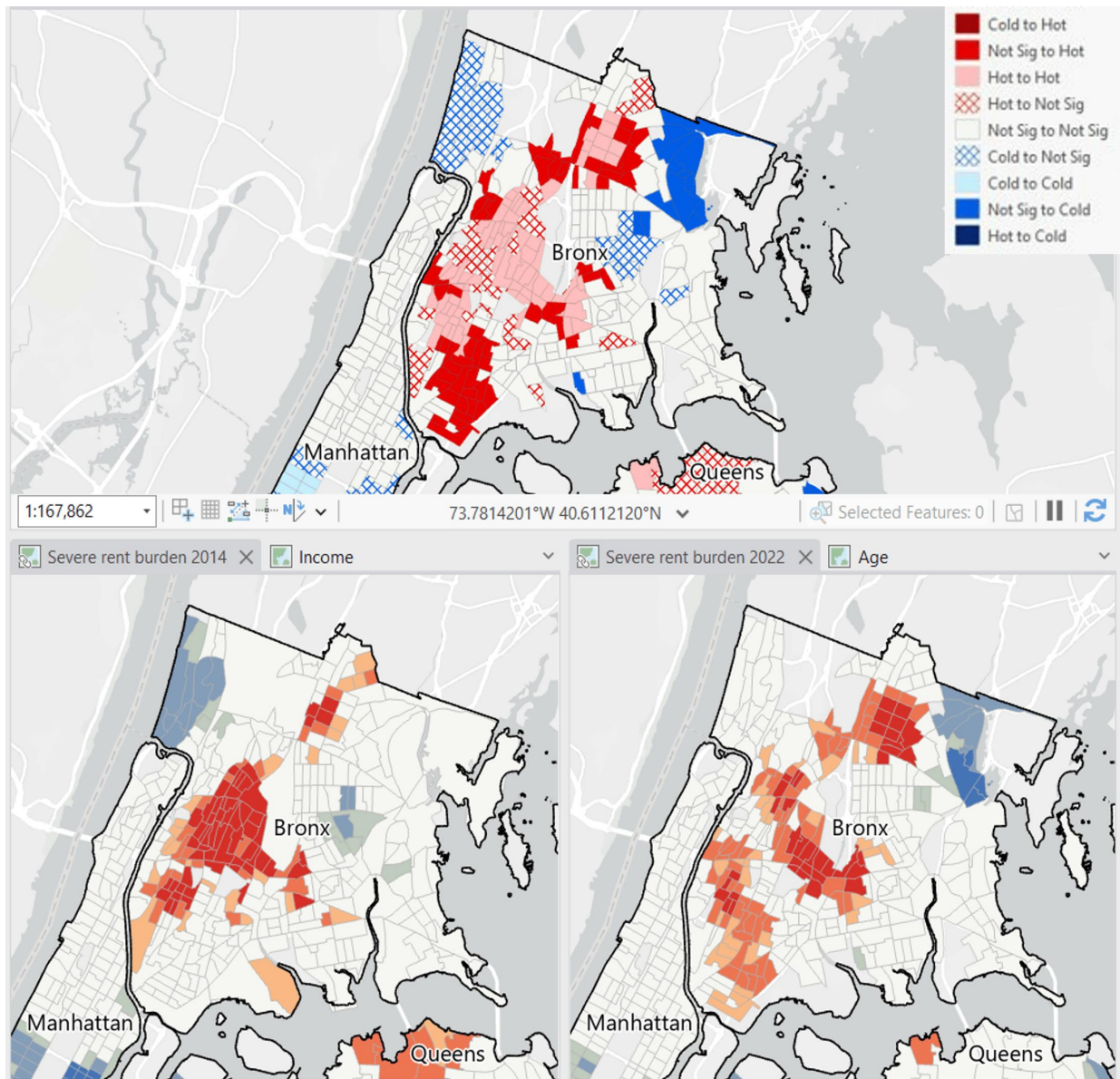


Figure 3-9: Digital twin data analytics using spatial statistics.²¹

3. Level 2 Capabilities for Data Analytics with Location may include the following. For illustrative purposes, the Level 2 capabilities shown in Figure 3-10 are also adapted from the UCGIS GIS&T Body of Knowledge, as introduced above, as well as the GISGeography Periodic Table for Spatial Analysis.²²

²¹ Esri ArcGIS Pro Introduction. Retrieved March 2024 from <https://www.esri.com/en-us/arcgis/products/arcgis-pro/>

²² GISGeography is a team of geographic information system (GIS) evangelists that provides resources for GIS professionals, researchers, and other interested parties. Their Periodic Table for Spatial Analysis is, like the UCGIS GIS&T BOK, an effort to describe the breadth of location analysis capabilities in a structured fashion. The Periodic Table for Spatial Analysis shares an affinity with the Digital Twin Capabilities Periodic Table due to shared and common use of this reference framework approach. See <https://gisgeography.com/spatial-analysis-periodic-table/>.

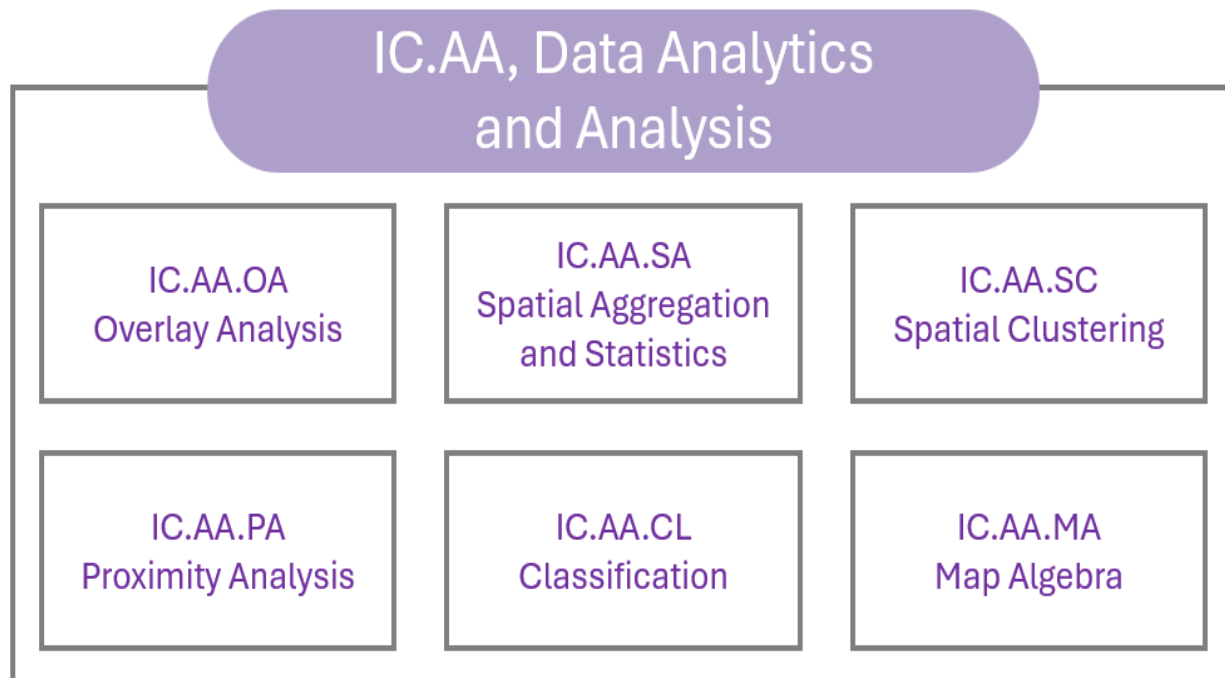


Figure 3-10: Level 2 capabilities for digital twin location-based data analysis.

These Level 2 capabilities might be described as follows:

“Our digital twin requires the ability to...”

- *IC.AA.OA, Overlay Analysis* – ... overlay two or more layers of map data to identify common areas where multiple conditions exist at the same location, based on overlapping map features,
- *IC.AA.PA, Proximity Analysis* – ... identify map features that are closest to each other, or to select features that are within a defined distance of other features,
- *IC.AA.SA, Spatial Aggregation and Statistics* – ... calculate statistics summarizing and characterizing a set of data within a given boundary or geographic area,
- *IC.AA.CL, Classification* – ... assign spatial data (assets or events shown on a map) to categories based on a defined range of values or characteristics,
- *IC.AA.SC, Spatial Clustering* – ...divide a set of data (assets or events shown on a map) into groups based on spatial proximity (e.g., assigning customers to a nearest store),
- *IC.AA.MA, Map Algebra* – ... generate new data values at a given location by performing mathematical functions on other data values located at the same location (typically for raster data such as images and other scanned data).

4 DESCRIBING LOCATION MODELS AND CHARACTERISTICS

Once the capabilities of a digital twin are selected and defined (as described in Chapter 3), organizations should then consider the characteristics of the locational data needed to deliver and use those capabilities.

Most digital twins – through their virtual representations – represent built assets, objects, and conditions in the real world.²³ These entities are commonly associated with location on the earth’s surface or relative to other entities. To help organizations identify the best approaches for capturing this locational information, this section describes digital twin spatial intelligence in terms of:

- Locational models – The different methods and approaches available to represent the location, shape, and spatial extent of entities (4.1)
- Locational characteristics – The key properties that should be considered and documented for the locational representation of real-world entities (4.2)

4.1 LOCATIONAL MODELS

Different locational data models can be used to represent real-world entities in the spatially intelligent digital twin. The model(s) selected depend on the level of fidelity and detail needed, and the capability requirements (see Chapter 3) that are to be supported. The three primary locational models include geometric/3D, spatial, and geospatial data:

1. Geometric/3D models – Geometric and 3D models are mathematical representations of an entity using points, lines, surfaces, and solids. In a digital twin, geometric / 3D models of built assets provide information about the shape and configuration of an asset but not about its location or spatial relationship to other assets or entities.

Geometric / 3D models are used in digital twins where the details of individual assets, such as a transformer bank in an electrical system, or a jet engine, must be visualized.

2. Spatial models – Spatial data pertains to features or phenomena distributed in three-dimensional space, including any space, not just the Earth’s surface. Spatial data describe the position and relationship between entities using a defined coordinate reference system. Spatial properties include the relative location, size, distance, connectivity, and distribution patterns in relationship to other entities. Spatial data models enable awareness of how objects are connected or otherwise locationally related to each other.

²³ Other digital twins may represent non-built assets such as financial information, human resources information, and the like. In the context of AECO, digital twins nearly always include representation of one or more built assets.

Spatially Intelligent Digital Twin Capabilities and Characteristics

Spatial models are used in digital twins where systems – and the relationships between items in the systems – must be visualized, such as a schematic layout portraying the connectivity of assets in an electrical grid, or to show the component items within a jet engine.

3. Geospatial models – Geospatial data are a subset of spatial data that pertains to locations on the Earth’s surface and near-surface environments, focusing on entities and phenomena tied to the Earth. Geospatial data are used for creating maps and visual representations that accurately represent the location of entities on the Earth’s surface.²⁴

Figure 4-1 illustrates the use of geometric/3D, spatial, and geospatial models to represent assets in an electrical power distribution system. Geometric/3D models are used to portray individual assets such as a transformer bank visually and accurately, spatial data models are used to portray a schematic of the distribution system, showing the connectivity of circuits in the system, and geospatial data models are used to accurately portray the location of assets in the system on a (map) view of the Earth’s surface.

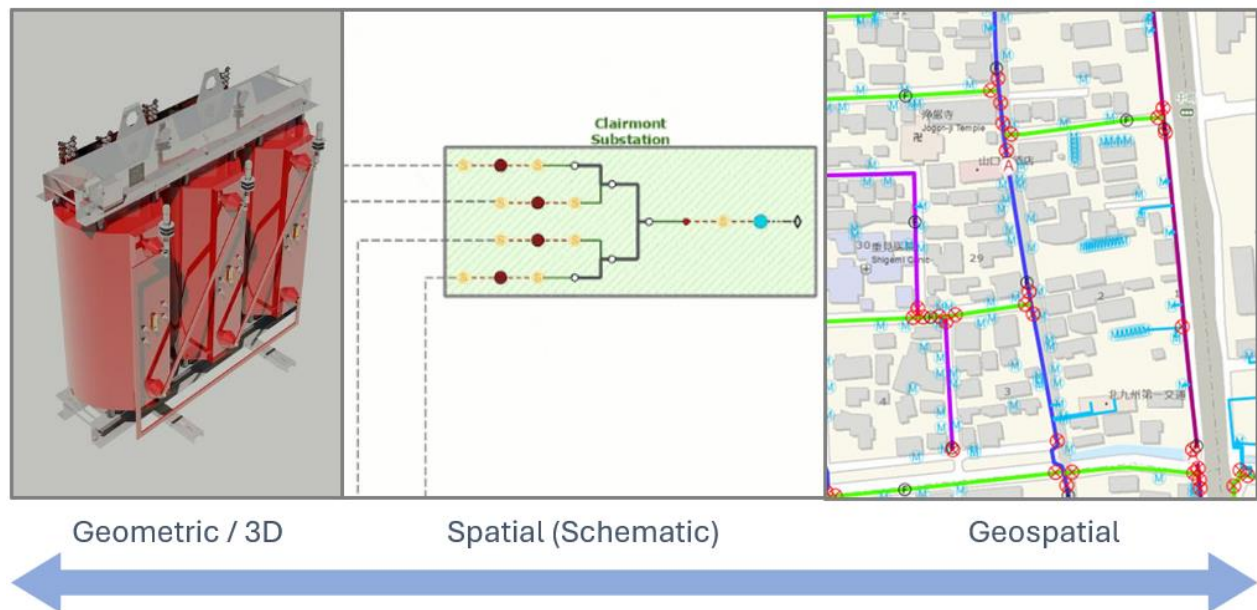


Figure 4-1: Spectrum of locational models.

SELECTING THE APPROPRIATE LOCATIONAL MODELS

Digital twins can use geometric/3D, spatial, and/or geospatial models to generate stored representations of real-world objects and assets.

More than one locational model may be used in each digital twin, and the stored representation of real-world entities may shift from geometric/3D to spatial (schematic) to geospatial (map) views as the user interacts with the digital twin system.

²⁴ Definition adapted from Microsoft Copilot generative AI, created on 12 Feb 2024.

Spatially Intelligent Digital Twin Capabilities and Characteristics

The applicability of each model will depend on the use cases and capability requirements (see Section 3) for the digital twin.

TECHNOLOGIES USED FOR EACH LOCATIONAL MODEL

The key technologies employed to create, manage, and use these locational data models include computer-aided drafting and design (CADD), building information modeling (BIM), and geographic information systems (GIS).

Presently, all three technology platforms support geometric/3D, spatial, and locational models. The selection of the appropriate locational technologies for use in a digital twin fall beyond the scope of this User's Guide. Many commercially available and open-source software products, services, and platforms provide CADD, BIM, and GIS capabilities for use within digital twin systems.

4.1.1 GEOMETRIC / 3D MODELS

Geometric/3D models are a mathematical representation of an object or entity using points, lines, surfaces, and solids. In a digital twin, 3D geometric models of built assets provide detailed and virtually realistic portrayals of the shape and configuration of an asset *but not about its location or spatial relationship to other assets or objects*.

3D modeling techniques include wireframing, solids modeling, surface modeling, and rendering of point clouds generated from reality capture²⁵ are used to generate digital representations of the outer surfaces, or envelope, as well as the interior configurations of real-world objects such as buildings or mechanical equipment items. Figure 4-2 shows a geometric (3D) model of a building.

²⁵ The Digital Twin Consortium has published several white papers on the topic of reality capture for digital twins. See <https://www.digitaltwinconsortium.org/publications>



Figure 4-2: Geometric (3D) model of a building in a digital twin.²⁶

EXTENDING GEOMETRIC/3D MODELS WITH LOCATION

A 3D model of an asset, such as a building as illustrated above, can be placed in its geographic context, showing the location of the asset. In this case, geometric/3D models are integrated with geospatial data, described further in Section 4.1.3.

However, this is not required. A digital twin may include the 3D model of an asset, such as a building, a transformer in an electrical system, a pump station, or a robot mechanism on an assembly line, etc. without any further locational information or context.

4.1.2 SPATIAL DATA MODELS

Spatial models extend geometry/3D models by describing the positioning and relationship between objects. Spatial properties include the location, size, distance, direction, connectivity, pattern, and movement in relationship to other objects.

Spatial models enable awareness of how objects are logically connected and enable capabilities such as tracing through a network, based on *topological* properties (which define how objects relate or connect).

Spatial properties such as connectivity can be shown in a schematic diagram, alternately referred to as a network diagram. Schematic diagrams are symbolic and simplified representations of the real world.

²⁶ Image courtesy of Motivf. See <https://www.motivf.com/>.

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Using spatial schematics and network diagrams, digital twins can simplify the virtual representation of an object or asset for easier understanding. For some digital twins – such as a digital twin representing a pump or a pump station – a schematic view may be preferred.²⁷

SPATIAL MODELS FOR INDIVIDUAL ASSETS

Figure 4-3 illustrates a schematic (network diagram) of the components of a pump. This schematic simplifies the great complexity of the pump itself, if viewed as an actual/accurate portrayal of the pump itself. Using a spatial model – a network diagram – of the pump in a digital twin may be sufficient, and ideal, to meet the use case requirements for a given digital twin.

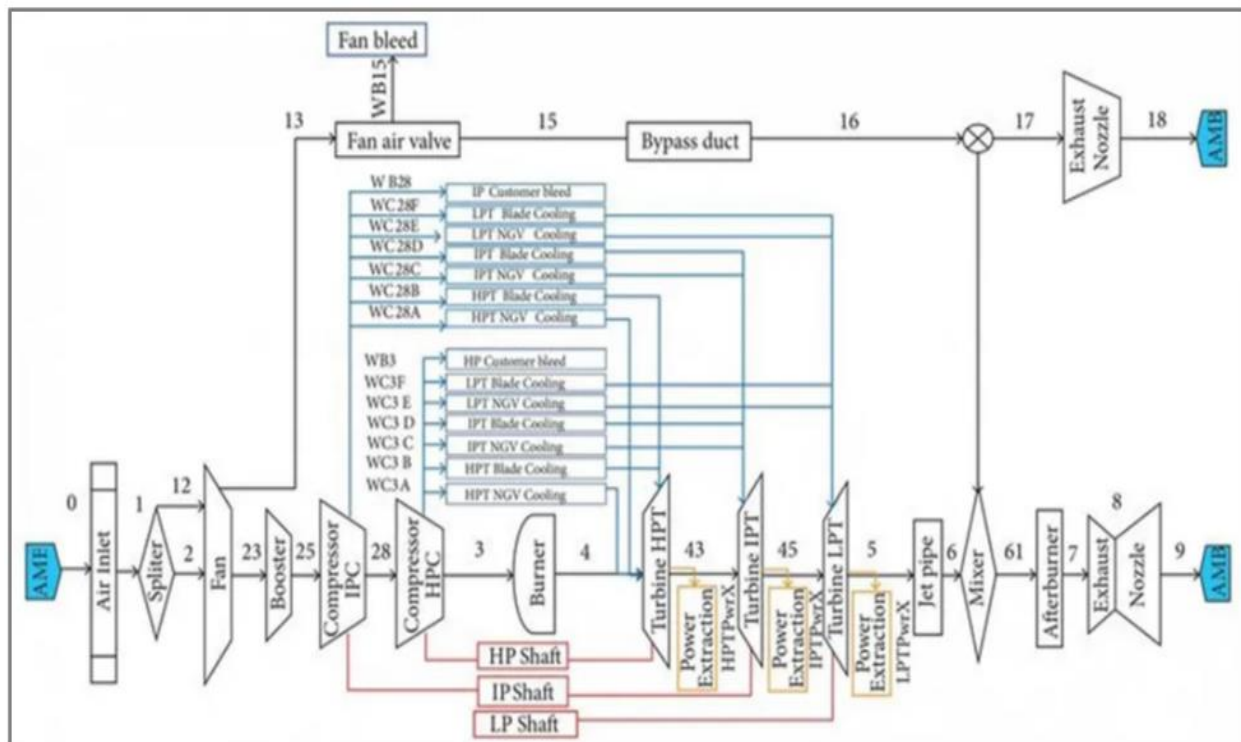


Figure 4-3: Spatial (schematic) representation of a single asset (pump).²⁸

SPATIAL MODELS FOR SYSTEMS

Spatial data models, including schematics and network diagrams, can also be used to portray simplified representations of complex systems with numerous assets, such as an electrical power distribution system.

²⁷ Derived from: <https://www.esri.com/arcgis-blog/products/utility-network/electric-gas/introduction-to-network-diagrams-what-they-are/>

²⁸ Image source: <https://www.pinterest.com/angelmadga/>

Spatially Intelligent Digital Twin Capabilities and Characteristics

For example, a simplified schematic network diagram allows for sufficient understanding of a transformer bank within an electrical distribution system, for the purposes of understanding operational dependencies and status, as shown in Figure 4-4.

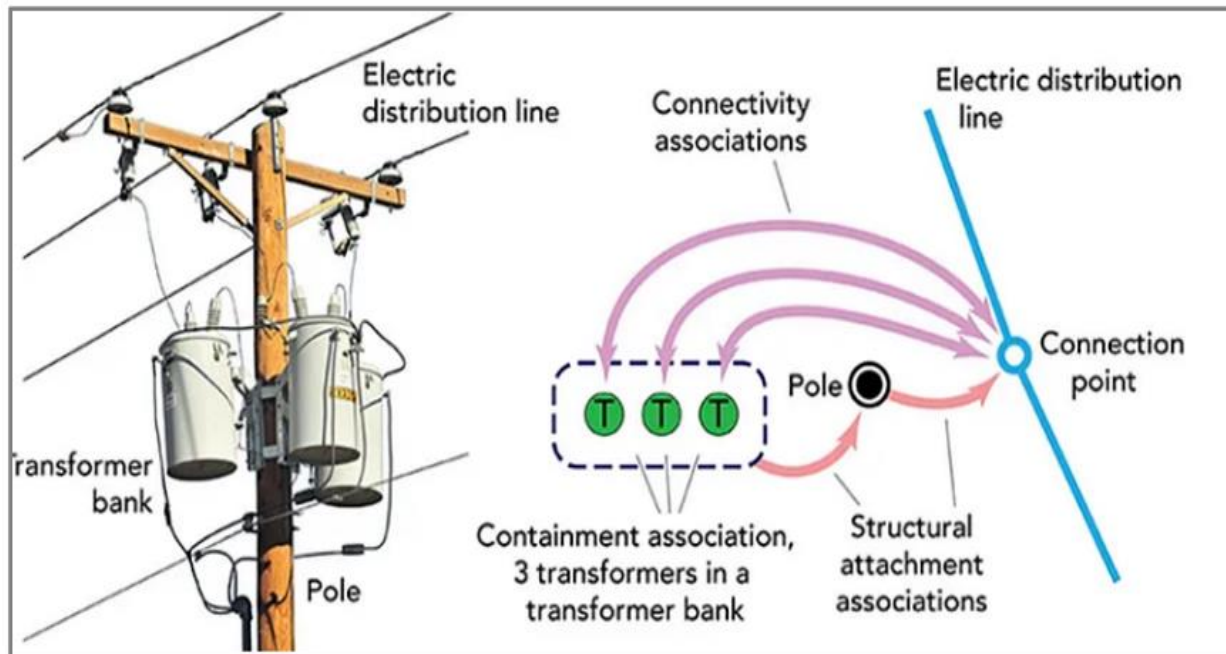


Figure 4-4: Spatial (Schematic) models to represent assets in a system.²⁹

This schematic view can be extended beyond the utility pole to the entire electrical distribution network grid. Figure 4-5 on the following page illustrates a logical representation of asset connectivity and hierarchy within the broader electrical grid using spatial properties.

This representation allows for an understanding of dependencies between assets in the network and enables modeling and analysis to understand the impacts of system outages.

The schematic of the electrical distribution network in Figure 4-5 is limiting in that it does not depict a locationally accurate view of where the assets are located, and as such does allow for understanding of the breadth of impact, who is impacted, and other systems that may be affected – this is where geospatial becomes important.

4.1.3 GEOSPATIAL DATA MODELS

Geospatial models enable the portrayal of the exact location and size of each asset on the earth's surface, building on the spatial properties of connectivity, overlap, etc. Geospatial data and representations provide a significantly more robust understanding of the system and offers a wealth of additional analytical opportunities.

²⁹ Source: Esri. "Vision and Overview of the Utility Network." 14 Feb 2020. <https://www.esri.com/arcgis-blog/products/utility-network/data-management/vision-and-overview-of-the-utility-network/>

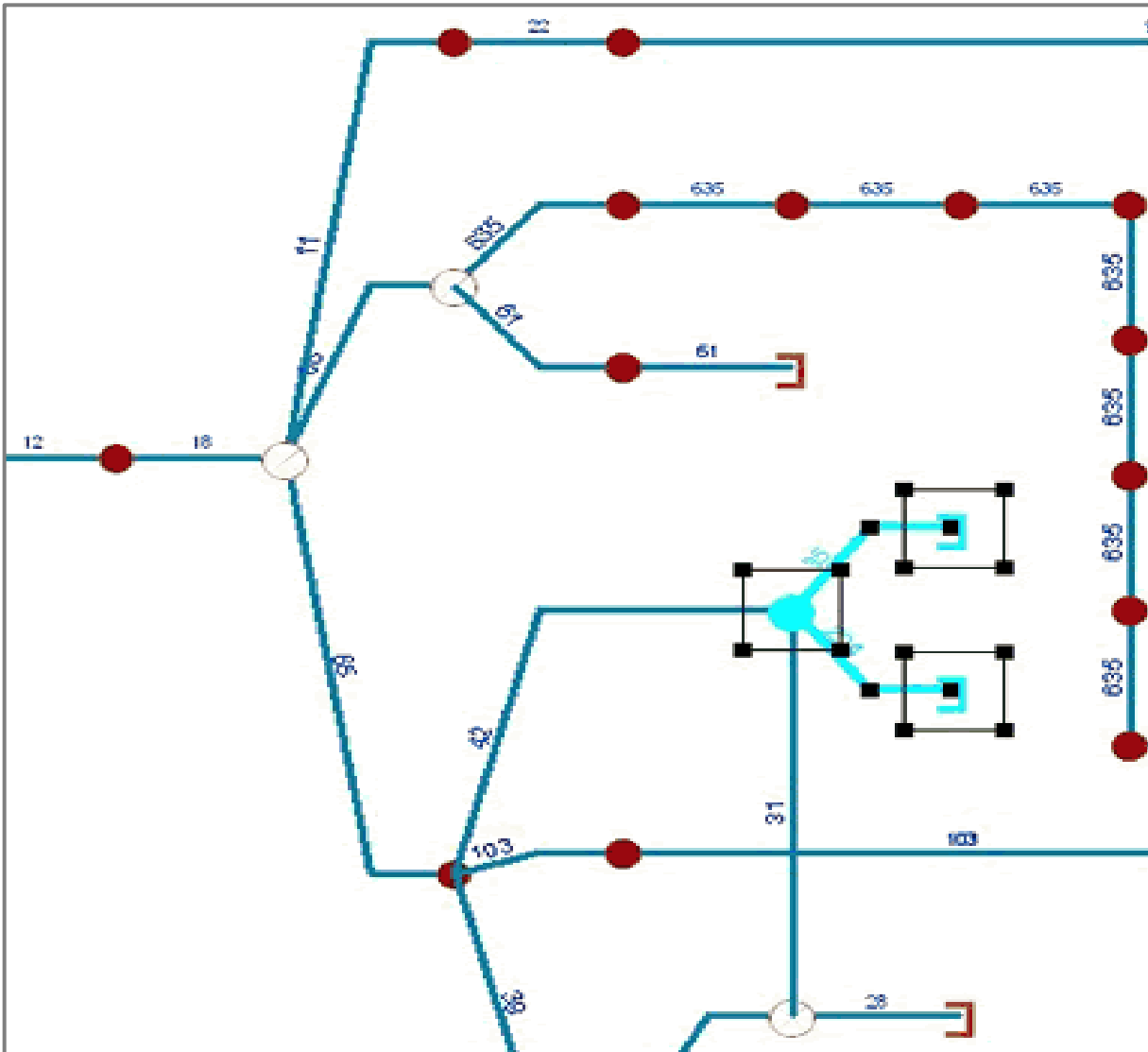


Figure 4-5: Spatial (schematic) representation of an electrical distribution system³⁰.

Figure 4-6 depicts a geospatial (map-based) representation of the same electrical distribution network portrayed in schematic form in Figure 4-5. This map view provides additional accuracy and context for digital twin users and provides a more accurate and precise understanding of the geographic extent of impacts of system operation.

³⁰ Image courtesy of Esri. See www.esri.com.

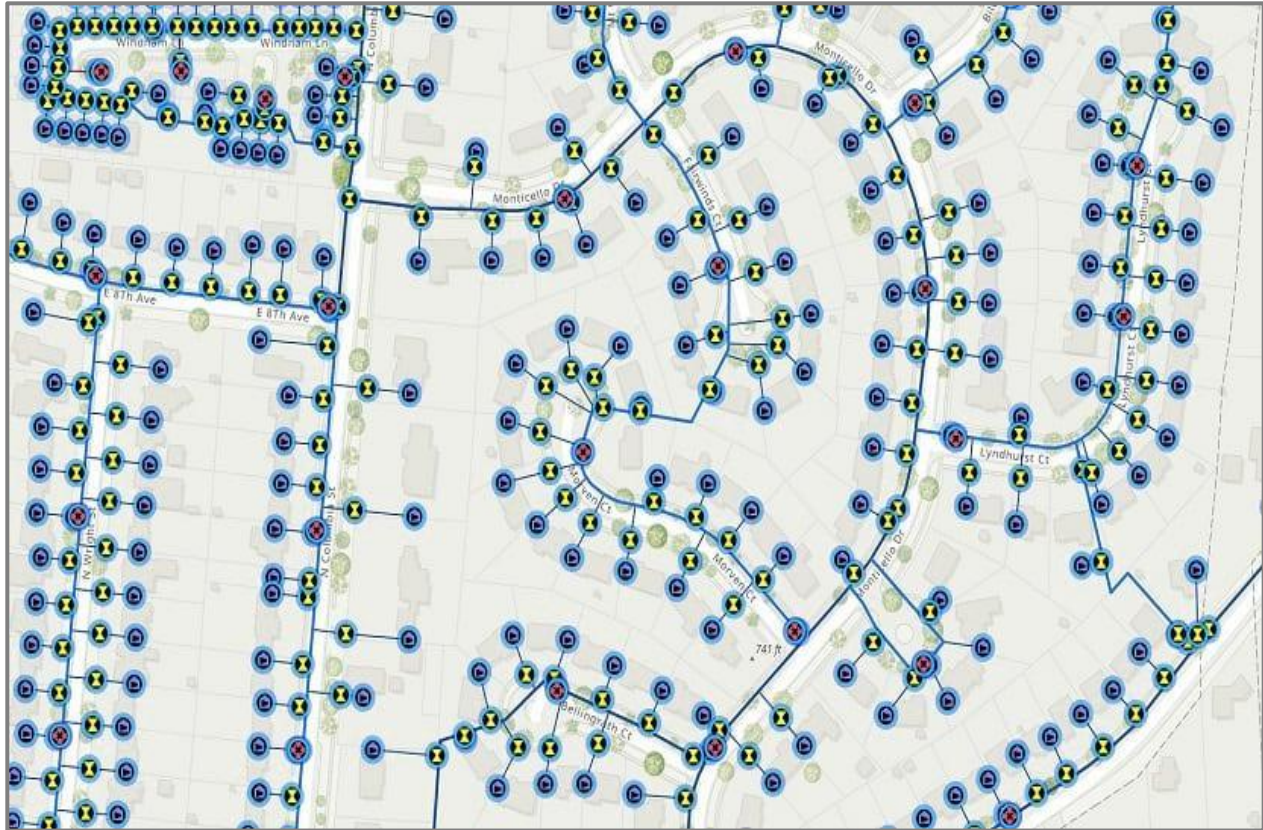


Figure 4-6: Geospatial representation of an electrical distribution system.³¹

4.1.4 COMBINING DATA MODELS FOR VIRTUAL REALITY (VR) CAPABILITIES

Digital twins may employ one or all the locational data models described above. As is increasingly common, CADD, BIM, and GIS technology solutions allow for integrated 3D virtual reality views of systems and regions. In these cases, geometry/3D and geospatial models are combined into a single view, or map display, to enable a comprehensive understanding of assets, systems, and their environment.

Figure 4-7 depicts geometric/3D models of individual buildings in a city, placed at their correct (accurate) geographic location, and overlaid with other geospatial data including a three-dimensional digital terrain model, and rendered with weather and other conditions to create a VR capability for use in a digital twin.

³¹ Nasser, Hussein. *Utility Network is now on ArcGIS Runtime 100.6*. (Aug 29, 2019). Esri ArcGIS Blog. <https://www.esri.com/arcgis-blog/products/utility-network/electric-gas/utility-network-is-now-on-arcgis-runtime-100-6/>.



Figure 4-7: Combining geometric/3D and geospatial models for virtual reality capabilities.³²

4.2 CHARACTERISTICS AND PROPERTIES OF LOCATIONAL DATA

Key characteristics (properties) of locational data that organizations should define include the following. These apply to all three locational data model types introduced in Section 4.1.

1. Spatial extent
2. Fidelity and level of development (LOD)
3. Locational reference (coordinate) systems
4. Spatial units
5. Map scale and measurement unity
6. Data types and data storage
7. Content organization (data models, schema, and standards)
8. License and access restrictions
9. Privacy and security

By selecting the locational model(s) and describing the corresponding key attributes of those locational models, organizations will then be able to describe their digital twin locational

³² Image source: <https://www.esri.com/news/releases/11-4qtr/esri-releases-new-version-of-cityengine.html>

capabilities in a detailed and structured fashion, in turn supporting digital twin design and solution development.

4.2.1 SPATIAL EXTENT

Spatial extent defines the locational limits of a project (or a digital twin's) data holdings, referred to as the area of interest (AOI) or project limits.

Understanding the spatial extent of a digital twin can help inform the type of locational models used (see 4.1) as well as the other characteristics and properties used for those locational data.

The spatial extent of a digital twin virtual representation may be:

1. Individual asset or facility – The digital twin and its related virtual representation is limited to a single asset (e.g., a pump in a pump station) or to a single building (e.g., a pump station).
2. Local – the digital twin and its related virtual representation represent a local area (a group of buildings, a neighborhood, campus, small town or city, county, small watershed, service area, or similar administrative unit or other geographic area).
3. Regional – the digital twin and its related virtual representation cover a large city, state, large watershed, or similarly large service area or administrative unit).
4. Global – the digital twin covers the entire globe, continent, or very large country such as Russia, Brazil, China, the United States, or similar.

METHODS TO DOCUMENT SPATIAL EXTENT

Spatial extent can be measured as a bounding box providing a set of geographic coordinates (see 4.2.3) within which the area of interest for a digital twin lies. Guidance is available from various organizations including:

1. International Standards Organization (ISO) Standard 19115-1:2014, Geographic Information Metadata – Defines the schema required for describing geographic information by means of metadata, including the extent of a geospatial data set, in addition to other properties and characteristics of the data set. See <https://www.iso.org/standard/53798.html>
2. NASA EarthDATA Wiki – Includes a set of Unified Metadata Models including a model to capture spatial extent of geographic data. See <https://wiki.earthdata.nasa.gov/display/CMR/Spatial+Extent>

4.2.2 FIDELITY AND LEVEL OF DEVELOPMENT (LOD)

ACCURACY AND PRECISION

Organizations should consider and document the accuracy, precision, and richness of descriptive information needed for their locational data in a digital twin³³. Several approaches are available to evaluate and communicate target levels of fidelity and detail for a digital twin virtual representation.

LEVELS OF DETAIL AND DEVELOPMENT

The BIM industry has defined and adopted two relevant measures: Levels of Detail (LOD) representing the amount of graphic detail a model contains, and Levels of Development (also LOD), representing the graphical detail as well as the amount of descriptive information (properties) the model contains.

While the LOD measures were developed for facility design, construction, and operations using BIM capabilities, these measures can be adopted and used to describe the level of detail for a digital twin's virtual representation, regardless of technologies used.

The Level of Development metric is best suited to describe the amount of detail contained in a digital twin virtual representation, when using the geometric/3D locational model described in Section 4.1.1.³⁴

1. LOD 100: Conceptual Design – A geometric/3D model of an asset represents the basic shape and configuration of the asset using generic representations, without detailed information, typically used to convey design intent.
2. LOD 200: Schematic Design – A model of an asset is more refined, incorporating enough detail to approximate quantities, sizes, shapes, orientation, and placement of elements of the model to evaluate design concepts.
3. LOD 300: Detailed Design – Similar to LOD 200 but with enough fidelity in the model to move from quantity (e.g., volume, size, cost) estimation to measurement, in enough detail to produce construction documents.
4. LOD 350: Construction Documentation – Models of assets include detailed information on assemblies, fabrication, and construction-level information, including interfaces between model elements and models of adjacent assets and structures.

³³ Methods for testing and measuring spatial and geospatial data accuracy fall beyond the scope of this User's Guide.

³⁴ Summary definitions are adapted from Autodesk (see <https://www.autodesk.com/solutions/bim-levels-of-development>) and BIMForum.org (see <https://bimforum.org/wp-content/uploads/2023/10/LOD-Spec-2023-Part-I-Public-Comment-Draft-2023-12-28.pdf>).

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5. LOD 400: Fabrication and Assembly – Models of assets are detailed enough to be used for fabrication and assembly purposes.
6. LOD 500: As-Built and Facility Management – Models of assets include information about installed and operational elements of the asset, reflecting real-world conditions for maintenance and facility management.

The virtual reality view of a city shown in Figure 4-7 is captured at Level of Development 200; sufficient to understand the overall design of buildings, based on approximated quantities, shapes, orientation, and placement.

Figure 4-8 illustrates the virtual representation of an elevator and elevator shaft at differing Levels of Development. Each LOD may be appropriate for use within a digital twin, given the use cases and capability requirements established for the digital twin.

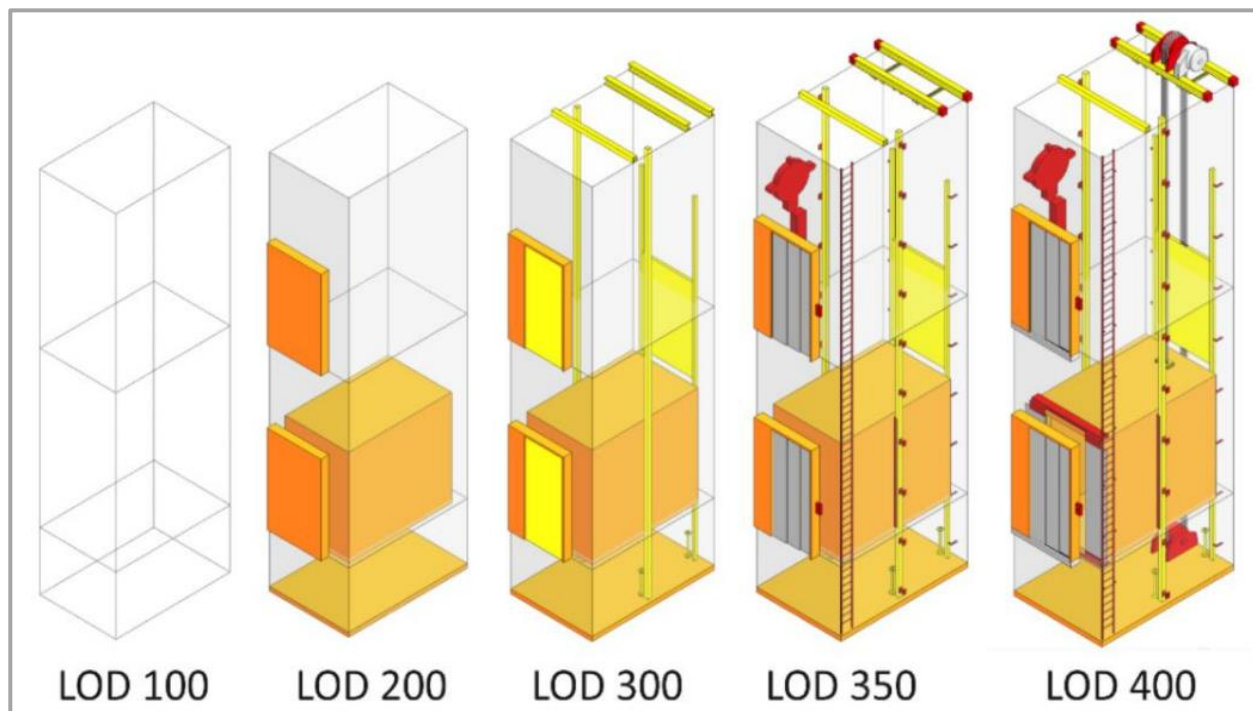


Figure 4-8: BIM levels of development (LOD).³⁵

As an alternative to adopting and using BIM LOD measures, organizations may describe the level of fidelity in their digital twin virtual representations – including both level of accuracy and level of descriptive attribution – using a generic set of measures:

³⁵ Source: Abualenien, Jimmy. Borrmann, Andre. “Levels of Detail, Development, Definition, and Information Need: A Critical Literature Review”. April 2022. Journal of Information Technology in Construction. Pg. 367. https://www.itcon.org/papers/2022_18-ITcon-Abualdenien.pdf

Spatially Intelligent Digital Twin Capabilities and Characteristics

1. Low fidelity – Relatively low and basic level of locational/geospatial level of information, basic attributes
2. Medium fidelity – Relatively moderate level of locational information that augments basic info with some selected additional attributes and details.
3. High fidelity – Relatively high and accurate level of locational information that incorporates as much as possible the ability to associate CPT elements with detailed geospatial attributes and characteristics.

4.2.3 LOCATIONAL REFERENCE (COORDINATE) SYSTEMS

Spatial reference systems (SRS), alternately referred to as coordinate reference systems (CRS), are mathematical frameworks to measure and capture locations on or near the Earth's surface.

Organizations should select and document the horizontal and vertical coordinate systems used to locationally reference the geometric/3D, spatial, and geospatial data in their digital twin virtual representations.

Thousands of coordinate systems are available; each of which is best suited to specific locales, regions, and data uses. While CADD, BIM, and GIS systems provide for data transformation between different coordinate systems and provide capabilities to integrate data stored in different coordinate systems into integrated views, organizations should identify the coordinate systems best suited to represent their location data, given their digital twin use cases, and spatial extent (see Section 4.2.1).

CLASSES OF COORDINATE SYSTEMS

1. Geographic coordinate systems – Coordinate systems that measure locations directly on the Earth's surface, such as latitude and longitude. These coordinate systems must mathematically account for irregularities in the spherical shape of the Earth, or the *geoid*.
2. Projected coordinate systems – Regional or global coordinate systems that portray features from the Earth's surface on a flat plane. As these coordinate systems project data from a spherical shape to a flat surface, project coordinate systems usually include high levels of distortion for some data being represented.
3. Cartesian coordinate systems – Highly localized, usually custom coordinate systems created for small areas where the curvature of the Earth is not of relevance. This type of coordinate system is best suited to digital twins focused on very small areas (e.g., a single building or asset) or for digital twin assets that may not be referenced to a location on the Earth's surface (e.g., a jet engine on an airplane).
4. Geocentric coordinate systems – Coordinate systems that measure locations and distances from the Earth's center of mass. Primarily used for high-altitude and near-earth orbit locational referencing.

Spatially Intelligent Digital Twin Capabilities and Characteristics

5. Datum – Coordinate systems must reference datum, which mathematically model the shape of the earth, referred to as the ellipsoid. Common datum includes the World Geodetic System (WGS) and North American Datum (NAD).

KEY STANDARDS AND GUIDELINES

1. Guide to Coordinate Reference Systems (CRS) Resources – Jointly published by the ISO, Open Geospatial Consortium (OGC) and International Association of Oil and Gas Producers (IOGP). See <https://committee.iso.org/files/live/sites/tc211/files/Resources/GuideToCRSRegistries3.pdf> (July 2022) and <https://www.iogp.org/wp-content/uploads/2019/09/373-07-02.pdf> (September 2019).
2. ISO Geodetic Registry (ISOGP) – The ISOGP is a registry of international coordinate reference systems and transformation between CRS. See <https://geodetic.isotc211.org/>.
3. ISO Standard 19111:2019, Geographic Information: Referencing by Coordinates – Guidance for how to select and capture coordinate system information, conformance guidelines, and UML schema for digital integration and discovery. See <https://www.iso.org/standard/74039.html>.
4. ISO Standard 6709:2022, Standard Representation of Geographic Point Location by Coordinates – Standards for the representation of latitude and longitude and other coordinate reference systems, including vertical measurements (height and depth). See <https://www.iso.org/standard/75147.html>.

4.2.4 MAP SCALE AND MEASUREMENT UNITS

MAP SCALE

Map scale is the ratio or proportional relationship between a map depiction of an entity and its actual size in the real world. A map scale of 1:24,000 means that one unit of measurement on the map equals 24,000 units of that same measure in the real world.

Map scales influence how users of a digital twin will perceive and understand the information being shown and can impact how users may make decisions based on information communicated through a map display.

Traditionally, printed maps would state the scale at which map features are rendered. For digital twins, an assumption is made that maps and locational data will be depicted through digital displays, and that digital twin users will be able to interact with the digital map to change display scales as needed (i.e., zooming in or out on the map display).

In digital twins, the map-based view of entities (assets and conditions) may include scale-dependent rendering. As digital twin users zoom in or out on the map display, the level of detail presented for those entities may change. As users “zoom out” (to show larger areas), the entities

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may be rendered at a lower level of detail. As users “zoom in” (to show smaller, more focused areas) the entities may be rendered at higher levels of detail.

For example, when viewing an elevator shaft in a digital twin of a building, zooming out to the full building may result in the elevator shaft being rendered at LOD 200 (see Figure 4-8). Conversely, zooming in on the elevator may result in that elevator shaft being rendered at LOD 400 (also see Figure 4-8).

Organizations should consider and document the following, to provide the best user experience (UX), to best convey the information in their digital twin, and to best meet defined digital twin capabilities (see Chapter 3):

1. Zoom levels – What standard (pre-defined) map scales should the digital twin map display support (should the digital twin limit how far the user can “zoom in” or “zoom out”).
2. Levels of detail – How much detail (LOD) should be rendered as digital twin users “zoom in” on the map display, and how much should entities be generalized as users “zoom out” on the map display.

COMBINING DATA AT DIFFERENT MAP SCALES

As organizations consider the spatial extent of their digital twin area of interest (see Section 4.2.1), source data at appropriate scales should be collected and compiled. A digital twin focused on a highly localized area of interest, such as a building, facility, or campus should avoid selecting and using data compiled and rendered at regional- and national-map scales, which will likely be less detailed than needed for the analysis and decision-making objectives of the digital twin. More fine-grained data, better fit for use, will be warranted to enable effective decision-making.

MEASUREMENT UNITS

Digital twin users may require the ability to measure units such as distance or area on the map display (this would be captured as part of the digital twin's locational capabilities as described in Section 3).

Organizations should consider and document what map units (linear feet, meters, etc.) their digital twins will offer, for measurements on the digital twin map display. The available units may be dependent on the coordinate systems selected for the digital twin's locational data (see Section 4.2.3). For example, spatial data using latitude/longitude (i.e., a global coordinate system) may not be able to support the ability to measure units in centimeters or inches.

Desired or target measurement units can be captured as a capability requirement, or as a property (characteristic) of spatial data, for documentation purposes.

4.2.5 DATA TYPES AND DATA STORAGE

DATA TYPES

Data can be categorized as structured, semi-structured, and unstructured. Locational data, as is most data in a digital twin context, is commonly stored as structured or semi-structured data:

1. **Structured** – Data that are organized into rows and columns, based on an agreed-upon logical categorization, data model, or “schema.” Structured data are typically stored and managed in spreadsheets and relational databases. Structured data are well-suited to quantitative analysis but are limited in that it can only be used for their intended purpose.

Spatial and geospatial data (see Sections 4.1.2 and 4.1.3) are typically stored as structured data.

2. **Unstructured** – Unstructured data are those data that are not organized by a logical structure, data model, or schema. Unstructured data include text, photos, video, audio files and in some cases, data streams from internet of things (IoT) sensors. Data resulting from reality capture scanning of build facilities – typically stored as “point cloud” data – are also unstructured.
3. **Semi-structured** – Data that have some structure, but without the constraints of a structured data model. Semi-structured data include extensible markup language (XML) and JavaScript Object Notation (JSON), both of which include defined tags that can be easily extended to include and capture additional information. Other semi-structured data include emails; the message within an email is typically unstructured text but the email contains defined structures to capture sender, recipient, date sent, etc.

Geometric/3D models (introduced in Section 4.1.1) are typically stored as semi-structured data.

Unless there is a specific capability enabled, or limited, by a given data type, there is little need to explicitly define or require data types in a digital twin requirements set.

Most technology solutions and software applications used for digital twins can store, serve, and use structured and semi-structured data. However, organizations should consider the data types used in their digital twins, to ensure that required capabilities can be delivered with the digital twin’s data assets.

DATA STORAGE

Locational data can be stored and accessed from a variety of formats, including files and data stored in relational and non-relational databases.

The methods for storing and retrieving locational data will be dictated by the technology architectures and the specific software solutions (commercial and/or open source CADD, BIM, GIS, database, etc.) used in a digital twin’s solution set.

4.2.6 CONTENT ORGANIZATION (DATA MODELS, SCHEMA, AND STANDARDS)

More important than data types and storage formats are the standards, data models, and database “schema” used to organize data in a digital twin.

Most data in a digital twin will be structured or semi-structured. In both cases, the logical organization of those data must be determined and well documented to enable data discovery, visibility, accessibility, understanding, and use.

This is the case for locational data as well. Spatial and geospatial data are typically organized into thematic layers; each layer representing a theme of data or a particular aspect of the real world. Within each layer, a set of descriptive attributes are defined to enable visualization and analysis of the data on each layer.

For instance, a digital twin locational data set may include a layer that represents parcels and land ownership - the top layer shown in Figure 4-9. Within the parcels layer, a set of attributes may be defined to capture land ownership of each parcel, as well as parcel type (public right-of-way, privately owned land, etc.), and other descriptive information. The parcel type is then used to render parcels on the map in different colors to enable data interpretation and understanding.

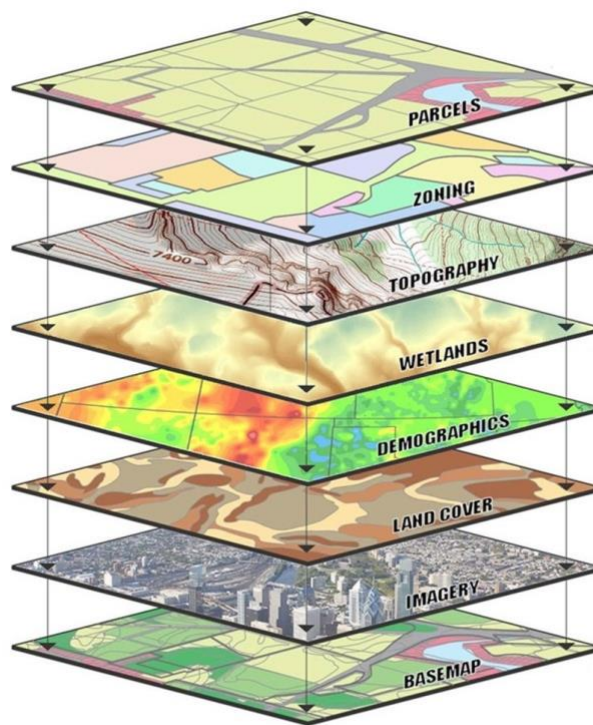


Figure 4-9: Geospatial data layers³⁶.

³⁶ United States Geological Survey. 24 Aug 2016. *GIS Data Layers*. Retrieved from <https://www.usgs.gov/media/images/gis-data-layers-visualization>

The following locational data content organization elements should be considered:

1. Data models – Data models provide a conceptual or logical organizing framework for data. A data model will define, for locational data, the data layers and layer characteristics as well as the descriptive attributes associated with each layer. Data models also describe the relationships between data in different layers. A data model is typically documented in a “database dictionary.”
2. Database schema – Database schema build on data models by providing the specific definitions needed to store and manage structured data in databases, as well as semi-structured data such as JSON and XML files. A schema will provide the additional specifications and information necessary to implement the data model.
3. Standards – A broad number of data models and schema are available to manage and use locational data. Data models and schema that are agreed-upon formally by a community or organization, or are widely adopted voluntarily across a given community, are typically published and made available as standards.

The range of standards available for locational data (geometric/3D models, spatial, and geospatial data) is large and the compilation of a comprehensive list of locational data standards is beyond the scope of this document.

Locational data standards (data models and schema) are available from government agencies, industry associations, and commercial entities, as described further, below.

Use of industry standards, rather than development of custom (bespoke, or “in-house”) standards, is recommended to best enable data sharing and interoperability between organizations and digital twin system-of-systems.

REPRESENTATIVE LOCATION-BASED (GEOSPATIAL) DATA STANDARDS

Government Standards³⁷

The United States National Geospatial Program (NGP) stewards The National Map, a national-scale geospatial database supporting a broad range of analytical and decision-making needs.³⁸ The National Map has defined a standard set of data layers, each layer with a defined set of content. For example, The National Map includes a hydrography data layer, and a set of descriptive attributes are established to organize and characterize hydrographic data in a

³⁷ International standards organizations such as the ISO typically do publish geospatial data content standards. However, ISO does not publish content standards defining how data should be logically organized, based on the subject matter being represented. Instead, ISO standards provide specifications for our spatial and geospatial data should be structured and organized at a more foundational level. ISO standards apply to geospatial data representing a broad range of subject matter and thematic focus.

³⁸ See <https://www.usgs.gov/ngp-standards-and-specifications/national-map-content>

standard fashion.³⁹ Digital twins requiring regional- or national-scale geospatial data of hydrographic features may wish to adopt, adapt, and use this standard.

Industry Standards

The Pipeline Open Data Standards, stewarded by the Pipeline Open Data Standard Association or PODS, publishes standards for pipeline and related infrastructure in the oil and gas industry. Organizations in this industry can consider industry consortia standards such as the PODS Pipeline Data Model⁴⁰ to organize geospatial representations of their pipeline assets in their digital twins.

As another example, the U.S. National BIM Standards Project Committee is a committee of the Institute of Building Sciences (NIBS) Facility Information Council (FIC) that stewards the National BIM Standard - United States (NBIMS-US). The NBIMS-US is a consensus-based data model facilitating the representation and exchange of building data for use in BIM systems. Organizations including locational (geometric/3D, spatial, and geospatial) representation of building and facility assets in their digital twins can consider adoption and use of this and related BIM standards.

DOCUMENTING DATA MODELS, SCHEMA, AND STANDARDS

Organizations should establish and fully document the data models and logical schema used for their digital twin's locational data. Additionally, if government, industry, or commercial standards are adopted and used as-is, or adapted and extended to meet the organization's digital twin capability requirements, this should be clearly stated and documented.

4.2.7 TEMPORAL AND TIME-BASED (SPATIAL-TEMPORAL) LOCATIONAL DATA

Digital twins monitor, portray, and manage assets in the real-world based on past, current, and predicted future conditions. The ability for digital twins to integrate and continually feed real- and near-real time data (from IOT sensors and other sources) into the digital twin virtual representation is a core digital twin capability.

Additionally, the assets monitored and managed in a digital twin's virtual representation may change over time. This includes the:

1. Asset locations – How the location of an asset (e.g., vehicles) may change over time,
2. Asset properties – How the asset's physical and related descriptive attributes may change over time (e.g., buildings and facilities that are restored, re-constructed, expanded, and improved),

³⁹ See <https://www.usgs.gov/ngp-standards-and-specifications/national-map-hydrography-content>

⁴⁰ See Pipeline Open Data Standards Association (Pods.org), <https://pods.org/data-models/> and <https://pods.org/pods-data-models/>

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3. Related data – How data associated with that asset, including asset operational performance, conditions, IOT sensor feeds, and the like, may change over time and be best presented via a map interface in a digital twin.

Digital twins should include the ability to manage spatial-temporal data. This can be defined in part through the selection and definition of digital twin capabilities (see Section 3). However, organizations should also consider and define the temporal characteristics of their locational data:

1. How should the digital twin's locational data be associated with time (date and time stamps)?
2. How can prior versions of locational data – representing past asset locations and conditions – be accessed and presented?
3. How can a digital twin's virtual representation (and associated system database) be designed to allow for viewing of temporal and time-series data, for data mining, historical trending, and analysis (prediction) of future conditions and performance?

4.2.8 LICENSE AND ACCESS RESTRICTIONS

Licensing and accessibility restrictions are key considerations when documenting the properties and characteristics of digital twin location data (as well as all data used within a digital twin).

Organizations should start by considering the breadth of users and stakeholders for their digital twins. What stakeholders (stakeholder roles, and/or named individuals) will be directly accessing, managing, and using digital twin data, and what stakeholders may be accessing derived products (summary data, maps, results of analytics performed in the digital twin, etc.)?

This then sets the foundation for consideration of data licensing and access restrictions. Organizations should consider the following when documenting the accessibility rights and restrictions of their digital twin data:

1. Consumed vs. published data – What locational data (reference maps, assets and their associated locations, IOT sensor feeds and their associated locations) are:
 - Consumed – Data loaded directly into a digital twin system database, accessed via web services from sources published by other entities, and then displayed and/or used for analytical purposes within the digital twin,
 - Published – Data that are created within the digital twin, including summary and derived data resulting from compilations of consumed source data, analytical functions within, and then displayed in or published as web services from the digital twin.
2. Open vs. proprietary data – Data consumed and published from a digital twin, including location data, may be either:

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- Open data – (Locational and other) data and content that can be freely accessed, used, modified, and shared for any purpose, subject to requirements that preserve provenance and openness⁴¹.
 - Proprietary (including commercial) data – Data that are restricted to defined uses and/or are accessible only by named organizations and individuals granted access rights and privileges.
3. Licensing – In the case of both open and proprietary data, if an organization intends to provide access to and/or share data from their digital twins, that data should be licensed and protected (including intellectual property, or IP, rights where applicable).
- Several organizations provide guidance on legal precedent and licensing approaches for both open and proprietary data including location-based data assets.^{42, 43}

4.2.9 PRIVACY AND SECURITY

Organizations should consider the implications on personal privacy, sensitive, and proprietary information when defining digital twin data (including locational data) assets. The introduction of location - the ability to associate information about people, assets, events, and conditions with a specific place, and to correlate information based on place - may lead to increased data privacy and security challenges.

Organizations should consider:

1. Privacy and security implications of their digital twin data – Including both data consumed by and published from the digital twin. This should include an assessment of the implications of depicting assets - including critical infrastructure, current and past asset conditions (including real-time IOT sensor feeds), and personally-identifying (PII) data – through a map interface and location intelligence capabilities.

Organizations such as but not limited to EthicalGEO, an initiative of the American Geographical Society, provide principles and guidance for the protection and ethical use of locational data and location-intelligence capabilities.⁴⁴

⁴¹ See <https://opendefinition.org/>

⁴² <https://opendefinition.org/guide/data/> is a good resource to start, including their detailed list of data licensing options for open data at <https://opendefinition.org/licenses/>

⁴³ See the July 2018 United Nations Committee of Experts on Global Geospatial Information Management (UN-GGIM) Compendium on Licensing of Geospatial Information: https://ggim.un.org/documents/E-C20-2018-9-Add_2-Compendium-on-Licensing-of-Geospatial-Information.pdf

⁴⁴ See <https://ethicalgeo.org/> and the March 2021 EthicalGEO LOCUS Charter at https://ethicalgeo.org/wp-content/uploads/2021/03/Locus_Charter_March21.pdf

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2. Organizational policies for data sensitivity and data sharing – What policies does the digital twin’s parent (hosting) organization have in place to categorize and control private and sensitive data.
3. Regional and governmental policies for data sensitivity and data sharing – Regulations such as the European Union (EU) General Data Protection Regulation (GDPR) and others such as but not limited to the Health Insurance Portability and Accountability Act (HIPAA), Federal Information Security Management Act (FISMA), California Consumer Privacy Act (CCPA) should be considered and used as guides to protect data used within and published from digital twins.

5 DOCUMENTING SPATIALLY INTELLIGENT DIGITAL TWIN LOCATIONAL CAPABILITIES AND CHARACTERISTICS

Spatially intelligent digital twin’s locational capabilities can be documented using approaches summarized in Section 3, and/or using other approaches. Many organizations have enterprise and business architectures and architecture frameworks that provide tools, templates, and guidance for defining and documenting business and technical capability requirements.

The characteristics and properties of a digital twin’s locational data may also be documented in a variety of formats and methods. Generally, the following options are available for documenting locational data characteristics and requirements:

1. Requirements documents – Functional requirements documents (FRDs) that capture and describe locational data (and other requirements) in narrative format with supporting tables, figures, and diagrams, Individual requirements can be captured as:
 - Declarative statements (e.g., “the digital twin shall include locational data captured at a horizontal positional accuracy of +/- ...”),
 - Use cases and agile user-stories (e.g., “as a digital twin user, I need to visualize the location of my assets at a spatial accuracy of +/-, so that I can [achieve a defined outcome]”), or as
 - Business rules (e.g., “our digital twins must be able to portray our assets at a spatial accuracy of +/- so that [a given business objective] can be met.”
2. Enterprise and/or business architecture artifacts – Architectural frameworks such as, but not limited to, the Zachman Framework for Enterprise Architecture, The Open Group Architecture Framework (TOGAF), the U.S. Federal Enterprise Architecture Framework (FEAF), the Business Architecture Guild’s Business Architecture Body of Knowledge (BIZBOK), and others provide guidance, tools, and templates (“artifacts”) to document capabilities and data requirements.

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3. JSON and XML schema and files – Extensible Markup Language (XML) and Javascript Object Notation (JSON) are both digital file formats that allow for the exchange of information between computer systems, applications, databases, and websites. Both are self-describing and human-readable, hierarchical (allowing nesting and association of data elements), and easily extensible and customizable.
 - Organizations can develop new, or adopt, adapt and extend existing XML and JSON schema to capture and define digital twin requirements and capabilities. Capturing digital twin capabilities and data requirements in this format enables digital (and system-to-system) discovery and access of digital twin requirements.
4. Digital geospatial metadata – Metadata are “data about data” or data files that describe other data assets. Two metadata formats (schema) are available to describe digital geospatial data:
 - ISO 19115 – an international geospatial metadata standard providing for the capture of locational data characteristics and properties including those introduced in Section 4.
 - U.S. Federal Geographic Data Committee (FFDC) Content Standards for Digital Geospatial Metadata (CSDGM) – An alternative metadata schema based heavily on ISO 19115 but tailored for U.S. federal geographic data. This is a legacy metadata schema and the FGDC now endorses the use of ISO metadata schema and standards.

Regardless of the format and approach used, documenting the Spatially Intelligent digital twin’s characteristics, including the location-related and location-intelligence capabilities of the twin, as well as the characteristics and properties of the location data that will enable those capabilities, is essential. Once documented, these requirements can then guide the design, development, testing, deployment and use of digital twins.

6 AUTHORS AND LEGAL NOTICE

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